

Sharp Maximum Estimates for Trace-Normalized Elliptic Operators

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Abstract

We study uniform L^p -to-supremum estimates for nondivergence-form elliptic operators with measurable symmetric coefficients, trace one, and a fixed positive lower eigenvalue bound. Explicit smooth radial examples establish the necessary exponent and exclude its endpoint in every dimension. A Mellin transform of the shifted determinant yields a positive-cone double-divergence estimate and proves sufficiency for $p > \max\{2, n/2, n[1 - (n-1)\delta]\}$. Together with the planar maximum theorem, this gives the exact threshold in every dimension other than three. All arguments retain the original interior C^2 and continuous zero-boundary function class. The three-dimensional results, their finite arithmetic certificates, and the precise remaining open range are presented at the end.

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1 The estimate and the scope of the results

Fix an integer $n \geq 2$ and a number $0 < \delta < 1/n$. We write

$$B_r = \{x \in \mathbb{R}^n : |x| < r\}, \quad a = 1 - (n-1)\delta, \quad P = P(n, \delta) = \max\{n/2, na\}.$$

Let Sym_n denote the real symmetric matrices, and put

$$\mathcal{K}_\delta = \{A \in \text{Sym}_n : A \geq \delta I, \text{ tr } A = 1\}.$$

An admissible coefficient field is a Lebesgue-measurable map $A : B_1 \rightarrow \text{Sym}_n$ belonging to $\mathcal{K}_\delta$ almost everywhere. For such a field define $L_A u = A : D^2 u = \sum_{i,j} a_{ij} \partial_i \partial_j u$. For $1 \leq p < \infty$, let $\text{MP}(n, \delta, p)$ mean that there is a finite constant $C = C(n, \delta, p) > 0$ such that

$$\max_{x \in \overline{B_1}} |u(x)| \leq C \left(\int_{B_1} |L_A u(x)|^p dx \right)^{1/p} \quad (1.1)$$

for every admissible A and every real-valued $u \in C^2(B_1) \cap C(\overline{B_1})$ with $u = 0$ on ∂B_1 and finite integral on the right. In particular, neither global $W^{2,p}$ regularity nor boundary differentiability is an assumption on u .

The question is whether

$$\text{MP}(n, \delta, p) \iff p > P(n, \delta). \quad (1.2)$$

Every inequality in this equivalence has the indicated strictness. The endpoint $p = P$ is part of the necessary direction, rather than an omitted case.

Theorem 1.1 (Dimensions other than three). *If $n = 2$ or $n \geq 4$, then (1.2) holds for every $0 < \delta < 1/n$ and every $1 \leq p < \infty$.*

The necessary direction is proved by explicit smooth radial functions in Theorem 2.1. Theorem 2.2 also resolves the radial restriction of the question in every dimension, while permitting arbitrary angular dependence of the coefficients. The planar sufficient direction uses a precisely stated theorem of Astala, Iwaniec and Martin [1]. In higher dimensions the main analytic estimate is

$$p > \max\{2, n/2, na\} \implies \text{MP}(n, \delta, p), \quad n \geq 3. \quad (1.3)$$

Its proof uses a Mellin integral of shifted determinant inequalities and an estimate for positive matrix-valued measures. The restriction $p > 2$ in (1.3) costs nothing when $n \geq 4$.

The exceptional dimension is treated only in the final mathematical section. There we give the additional proved ranges, distinguish the computer-assisted assertions from the analytic ones, and state exactly what remains unproved. This manuscript does not assert (1.2) in its full unrestricted form.

Proof conventions and external inputs. The project-specific arguments are given below without reference to research notebooks. We use classical L^p boundedness of the Hessian Riesz transform, smooth elliptic Dirichlet solvability and the classical maximum principle, and standard measure representation and interpolation theorems. The less routine external inputs—periodic Monge–Ampère solvability, the planar maximum theorem, the planar Burkholder integral inequality, and the Hilbert-valued martingale inequality—are each stated in the form used and attributed. Finite arithmetic certificates used in the last section are part of the accompanying reproducibility supplement; the manuscript specifies their exact assertions, verification rules and domains. No numerical experiment is used as a substitute for a sign certificate.

Notation. For matrices, $|H|$ is the Frobenius norm and $\|H\|_{\text{op}}$ the operator norm; $A : H = \text{tr}(AH)$. The distribution $\text{Div}^2 M$ is defined by $\langle \text{Div}^2 M, \varphi \rangle = \int D^2 \varphi : dM$. The total variation of a scalar signed measure ν is $\|\nu\|_{\text{TV}}$. All unspecified constants are finite and may depend on the displayed fixed parameters, but never on the particular admissible coefficient field or function.

Lemma 1.2 (Elementary coefficient facts). *Every $A \in \mathcal{K}_\delta$ has its eigenvalues in $[\delta, a]$. For $H \in \text{Sym}_n$, the set of possible contractions $A : H$ is exactly*

$$\{A : H : A \in \mathcal{K}_\delta\} = [\delta \text{tr} H + (1 - n\delta)\lambda_{\min}(H), \delta \text{tr} H + (1 - n\delta)\lambda_{\max}(H)]. \quad (1.4)$$

Furthermore, if an estimate holds at p_0 , it holds at every finite $p \geq p_0$; if it holds for δ_0 , it holds for every $\delta \geq \delta_0$ with $\delta < 1/n$.

Proof. The eigenvalue upper bound follows by subtracting the other $n - 1$ eigenvalues from the trace. Write $A = \delta I + (1 - n\delta)Q$, where $Q \geq 0$ and $\text{tr} Q = 1$. The extrema of $Q : H$ are the extreme eigenvalues of H , attained by rank-one spectral projections, and every intermediate value is attained by convex combinations. This proves (1.4). The last two assertions follow from $\|f\|_{p_0} \leq |B_1|^{1/p_0 - 1/p} \|f\|_p$ and $\mathcal{K}_\delta \subset \mathcal{K}_{\delta_0}$, respectively. $\square$

2 Sharp necessity and the radial problem

Throughout this section, $a = 1 - (n - 1)\delta$ and $P = \max\{n/2, na\}$, as above. We write $\omega_{n-1} = |S^{n-1}|$.

Theorem 2.1 (Necessity, including the endpoint). *For every $n \geq 2$ and $0 < \delta < 1/n$, there is a single admissible coefficient field A_* , depending only on n and δ , with the following property. For every $1 \leq p \leq P$, there is a sequence of nonnegative radial functions*

$$u_j \in C^\infty(B_1) \cap C(\overline{B_1}), \quad u_j|_{\partial B_1} = 0,$$

whose source norms are finite and nonzero and for which

$$\frac{\max_{\overline{B_1}} |u_j|}{\|L_{A_*} u_j\|_{L^p(B_1)}} \longrightarrow \infty.$$

In particular, the estimate in the target problem can hold only if $p > P$.

Proof. Set

$$d = \frac{1}{a}, \quad \vartheta = \min\{2, d\}.$$

Since $1/n < a < 1$, we have $1 < d < n$, and hence $P = n/\vartheta > 1$. Define the fixed field by

$$A_*(x) = \delta I + (1 - n\delta) \frac{x \otimes x}{|x|^2} \quad (x \neq 0), \quad A_*(0) = I/n. \quad (2.1)$$

This field is measurable and symmetric. Away from zero, its radial eigenvalue is a and its $n - 1$ tangential eigenvalues are δ ; at zero all its eigenvalues equal $1/n$. It therefore has trace one and satisfies the required lower ellipticity bound at every point. For a radial function $u(x) = U(r)$, $r = |x| > 0$, we have

$$L_{A_*} u = aU'' + (1 - a) \frac{U'}{r} = a \left(U'' + (d - 1) \frac{U'}{r} \right). \quad (2.2)$$

For any nonzero, nonnegative $f \in C_c^\infty((0, 1))$, put

$$U_f(r) = \frac{1}{a} \int_r^1 t^{1-d} \int_0^t s^{d-1} f(s) \, ds \, dt, \quad 0 \leq r \leq 1, \quad u_f(x) = U_f(|x|). \quad (2.3)$$

The inner integral vanishes near zero. Thus U_f is constant near zero, and u_f is smooth throughout B_1 , including the origin. It is continuous on the closed ball, vanishes on its boundary, and is nonnegative and radially nonincreasing. Differentiating (2.3) and using (2.2) gives

$$-L_{A_*} u_f(x) = f(|x|) \quad \text{for every } x \in B_1, \quad (2.4)$$

where f is extended by zero at the origin. Indeed, both sides vanish there because u_f is constant in a neighborhood of zero. In particular, every one of these functions has a finite source norm.

Tonelli's theorem yields

$$u_f(0) = \int_0^1 f(s) W(s) s^{n-1} \, ds, \quad W(s) = \frac{1}{a} s^{d-n} \int_s^1 t^{1-d} \, dt. \quad (2.5)$$

More explicitly,

$$W(s) = \begin{cases} \frac{s^{2-n}(1 - s^{d-2})}{a(d-2)}, & d > 2, \\ \frac{1}{a} s^{2-n} \log(1/s), & d = 2, \\ \frac{s^{d-n}(1 - s^{2-d})}{a(2-d)}, & d < 2. \end{cases}$$

Consequently, with $r_0 = 1/4$, there is a constant $c > 0$, depending only on n and δ , such that

$$W(s) \geq cs^{\vartheta-n} \quad (0 < s \leq r_0). \quad (2.6)$$

First suppose $1 \leq p < P$. Fix a nonzero nonnegative function $\varphi \in C_c^\infty((1, 2))$ and, for $0 < \varepsilon < r_0/2$, set

$$f_\varepsilon(s) = \varepsilon^{-n/p} \varphi(s/\varepsilon).$$

Polar integration and (2.4) give

$$\|L_{A_*} u_{f_\varepsilon}\|_{L^p(B_1)}^p = \omega_{n-1} \int_1^2 \varphi(t)^p t^{n-1} \, dt,$$

a positive number independent of ε . On the other hand, (2.5)–(2.6) imply

$$\max_{B_1} |u_{f_\varepsilon}| = u_{f_\varepsilon}(0) \geq c\varepsilon^{\vartheta-n/p} \int_1^2 \varphi(t) t^{\vartheta-1} \, dt \longrightarrow \infty,$$

because $p < n/\vartheta$.

It remains to treat $p = P$. Choose smooth functions $\eta, \zeta : [0, \infty) \rightarrow [0, 1]$ satisfying

$$\eta = 0 \text{ on } [0, 1], \quad \eta = 1 \text{ on } [2, \infty), \quad \zeta = 1 \text{ on } [0, r_0/2], \quad \zeta = 0 \text{ on } [r_0, \infty).$$

For $0 < \varepsilon < r_0/4$, let

$$f_\varepsilon(s) = s^{-\vartheta} \eta(s/\varepsilon) \zeta(s).$$

For each fixed ε , this is a nonnegative member of $C_c^\infty((0, 1))$, so (2.3) remains an admissible smooth test function. Since $\vartheta P = n$,

$$\|L_{A*} u_{f_\varepsilon}\|_{L^P(B_1)}^P \leq \omega_{n-1} \int_\varepsilon^{r_0} \frac{ds}{s} = \omega_{n-1} \log(r_0/\varepsilon),$$

whereas

$$u_{f_\varepsilon}(0) \geq c \int_{2\varepsilon}^{r_0/2} \frac{ds}{s} = c \log \frac{r_0}{4\varepsilon}.$$

Their ratio therefore diverges, since $P > 1$. This argument also includes $d = 2$, when the exact weight has the additional positive logarithmic factor. Taking any sequence $\varepsilon_j \downarrow 0$ proves the assertion in every case. All coefficient hypotheses, regularity conditions, and source integrability requirements hold for every individual member of the sequence, before taking a limit. $\square$

Theorem 2.2 (Complete characterization for radial functions). *Fix $n \geq 2$, $0 < \delta < 1/n$, and $1 \leq p < \infty$. There is a finite constant $C = C(n, \delta, p)$ such that*

$$\max_{\overline{B_1}} |u| \leq C \|L_A u\|_{L^p(B_1)}$$

for every admissible coefficient field A and every radial function $u \in C^2(B_1) \cap C(\overline{B_1})$ with zero boundary values and $L_A u \in L^p(B_1)$ if and only if $p > P$. No radial symmetry of A is assumed.

Proof. Necessity follows from Theorem 2.1, whose examples are radial and use the one fixed coefficient field (2.1). We prove sufficiency for $p > P$.

Write $u(x) = U(|x|)$ and $f = L_A u$. For $r > 0$ and $\theta \in S^{n-1}$,

$$D^2 u(r\theta) = U''(r) \theta \otimes \theta + \frac{U'(r)}{r} (I - \theta \otimes \theta).$$

Every eigenvalue of A lies in $[\delta, a]$ almost everywhere: the upper bound follows by subtracting the other $n - 1$ eigenvalues from its trace. Thus

$$b(r, \theta) = \theta^T A(r\theta) \theta \in [\delta, a]$$

almost everywhere, and the trace normalization gives

$$f(r\theta) = b(r, \theta) U''(r) + (1 - b(r, \theta)) \frac{U'(r)}{r}.$$

For almost every $r \in (0, 1)$, define the spherical averages

$$B(r) = \frac{1}{\omega_{n-1}} \int_{S^{n-1}} b(r, \theta) \, d\sigma(\theta), \quad F(r) = \frac{1}{\omega_{n-1}} \int_{S^{n-1}} f(r\theta) \, d\sigma(\theta).$$

Fubini's theorem provides measurable versions, and we may assign arbitrary values in $[\delta, a]$ to B on its exceptional set. We have

$$\delta \leq B \leq a, \quad BU'' + (1 - B)\frac{U'}{r} = F \quad \text{almost everywhere on } (0, 1). \quad (2.7)$$

Moreover, Jensen's inequality and polar integration imply

$$\left(\int_0^1 |F(r)|^p r^{n-1} dr \right)^{1/p} \leq \omega_{n-1}^{-1/p} \|f\|_{L^p(B_1)}. \quad (2.8)$$

Put

$$\kappa = \frac{1-a}{a} > 0, \quad c(r) = \frac{1-B(r)}{B(r)}, \quad \mu(r) = \exp\left(-\int_r^1 c(t) \frac{dt}{t}\right).$$

Then $\kappa \leq c \leq (1-\delta)/\delta$. The function μ is positive and locally absolutely continuous on $(0, 1)$, with $\mu' = c\mu/r$ almost everywhere. Equation (2.7) therefore becomes

$$(\mu U')' = \mu F/B \quad \text{almost everywhere on } (0, 1).$$

Radial C^2 regularity at the origin implies $U'(r) = O(r)$. Also $\mu(r) \leq r^\kappa$, so $\mu(r)U'(r) \rightarrow 0$ as $r \downarrow 0$. The Hessian of u is bounded near the origin, and positivity and trace one of A imply that f , and hence F , is essentially bounded there. Thus $\mu F/B$ is integrable near zero, and integration from zero is justified. Dividing by $\mu(r)$ gives

$$U'(r) = \int_0^r \frac{F(s)}{B(s)} \exp\left(-\int_s^r c(t) \frac{dt}{t}\right) ds, \quad 0 < r < 1.$$

In particular,

$$|U'(r)| \leq \delta^{-1} \int_0^r |F(s)|(s/r)^\kappa ds. \quad (2.9)$$

This step uses only the measurable bounds on B , and consequently allows arbitrary angular dependence and radial oscillation of A .

Define the nonnegative kernel

$$w_\kappa(s) = s^\kappa \int_s^1 r^{-\kappa} dr = \begin{cases} \frac{s^\kappa - s}{1 - \kappa}, & \kappa \neq 1, \\ s \log(1/s), & \kappa = 1. \end{cases}$$

Tonelli's theorem applied to (2.9) yields

$$\int_0^1 |U'(r)| dr \leq \delta^{-1} \int_0^1 |F(s)| w_\kappa(s) ds. \quad (2.10)$$

Since $p > P > 1$, let $p' = p/(p-1)$ and set

$$H_{n,\kappa,p} = \left(\int_0^1 w_\kappa(s)^{p'} s^{-(n-1)(p'-1)} ds \right)^{1/p'}.$$

This number is finite. Indeed, near zero, w_κ is comparable to s^κ if $\kappa < 1$, to s if $\kappa > 1$, and equals $s \log(1/s)$ if $\kappa = 1$. For a power s^γ , the displayed integrability condition is

$$\gamma p' - (n-1)(p'-1) > -1, \quad \text{equivalently} \quad p > \frac{n}{1+\gamma}.$$

It therefore requires $p > n/(1 + \kappa) = na$ when $\kappa < 1$ and $p > n/2$ when $\kappa \geq 1$. In the case $\kappa = 1$, the logarithmic factor is integrable above this threshold and makes the integral diverge at equality. These conditions are exactly $p > P$. Near one, $w_\kappa(s)$ is comparable to $1 - s$, so there is no additional restriction.

Weighted Hölder's inequality, (2.8), and (2.10) now give

$$\int_0^1 |U'(r)| \, dr \leq \delta^{-1} H_{n,\kappa,p} \omega_{n-1}^{-1/p} \|L_A u\|_{L^p(B_1)}.$$

In particular U' is integrable up to the boundary; this conclusion has not been assumed. For $0 \leq r < R < 1$, the fundamental theorem of calculus gives $U(R) - U(r) = \int_r^R U'(t) \, dt$. Letting $R \uparrow 1$ and using only continuity of U at one and $U(1) = 0$, we obtain

$$\max_{\overline{B_1}} |u| \leq \int_0^1 |U'(r)| \, dr \leq \delta^{-1} \omega_{n-1}^{-1/p} H_{n,\kappa,p} \|L_A u\|_{L^p(B_1)}.$$

The constant depends only on n , δ , and p , as required. $\square$

3 The complete planar case

The only deep input in this section is the planar maximum estimate of Astala, Iwaniec, and Martin. We state precisely the form used here. This is an external theorem; the normalization and passage to the regularity class of the present problem are proved below.

Theorem 3.1 (Astala–Iwaniec–Martin, determinant-one form). *Let $\Omega \subset \mathbb{R}^2$ be a bounded domain, and let $B : \Omega \rightarrow \mathbb{R}^{2 \times 2}$ be a Lebesgue-measurable symmetric positive definite matrix field. Suppose, almost everywhere, that*

$$\det B = 1, \quad \frac{\lambda_{\max}(B)}{\lambda_{\min}(B)} \leq K, \quad 1 \leq K < \infty.$$

If $p > 2K/(K + 1)$, then every real $v \in W^{2,p}(\Omega)$ and every relatively compact subdomain $D \Subset \Omega$ satisfy

$$\max_{\overline{D}} |v| \leq \max_{\partial D} |v| + C(K, p) (\text{diam } \Omega)^{2-2/p} \|B : D^2 v\|_{L^p(\Omega)}. \quad (3.1)$$

The constant is uniform over these fields B and functions v .

This is Theorem 3.5, equations (17)–(18), of [1], restricted to determinant-one matrices. In that paper the distortion K_B is the largest-to-smallest eigenvalue ratio, and the coefficient normalization is $K_B^{-1/2} I \leq B \leq K_B^{1/2} I$; see its equations (7), (9), and (10). Thus the parameter K in Theorem 3.1 is the eigenvalue ratio, not its square root. The theorem imposes no upper restriction on p . We have included the fixed distortion parameter in the notation for the constant. The continuous representative on $\overline{D}$ exists because $p > 1$ and $D \Subset \Omega$.

Theorem 3.2 (Sharp planar estimate). *Fix $0 < \delta < 1/2$ and $1 \leq p < \infty$. A finite constant $C = C(\delta, p)$ such that*

$$\max_{\overline{B_1}} |u| \leq C \|A : D^2 u\|_{L^p(B_1)}$$

for every Lebesgue-measurable symmetric A satisfying $\text{tr } A = 1$ and $A \geq \delta I$ almost everywhere, and every real $u \in C(\overline{B_1}) \cap C^2(B_1)$ with $u = 0$ on ∂B_1 and $A : D^2 u \in L^p(B_1)$, exists if and only if

$$p > 2(1 - \delta).$$

Proof. The necessity, including failure at equality, follows from Theorem 2.1, since $\max\{1, 2(1-\delta)\} = 2(1-\delta)$. We prove sufficiency without assuming that u belongs to $W^{2,p}(B_1)$.

Suppose $p > 2(1-\delta)$ and put

$$K = \frac{1-\delta}{\delta}, \quad \tilde{A}(x) = \frac{A(x)}{\sqrt{\det A(x)}}.$$

Any null set on which the coefficient hypotheses fail may first be filled with $A = I/2$. This preserves measurability and the source norm. The eigenvalues of A lie in $[\delta, 1-\delta]$ and sum to one. Consequently

$$\det A \geq \delta(1-\delta), \quad \frac{\lambda_{\max}(\tilde{A})}{\lambda_{\min}(\tilde{A})} = \frac{\lambda_{\max}(A)}{\lambda_{\min}(A)} \leq K, \quad \det \tilde{A} = 1. \quad (3.2)$$

The normalized field therefore satisfies the hypotheses of Theorem 3.1. Moreover,

$$|\tilde{A} : D^2 u| \leq [\delta(1-\delta)]^{-1/2} |A : D^2 u|, \quad \frac{2K}{K+1} = 2(1-\delta) < p. \quad (3.3)$$

For $0 < r < R < 1$, set

$$v_R(y) = u(Ry), \quad B_R(y) = \tilde{A}(Ry), \quad y \in B_1.$$

Because $\overline{B_R}$ has a neighborhood compactly contained in B_1 , the function u and its derivatives through order two are bounded on $\overline{B_R}$. Hence $v_R \in W^{2,p}(B_1)$, even though no global Sobolev assumption on u was made. The chain rule and the change of variables $x = Ry$ give the exact scaling identity

$$\|B_R : D_y^2 v_R\|_{L^p(B_1)} = R^{2-2/p} \|\tilde{A} : D^2 u\|_{L^p(B_R)}. \quad (3.4)$$

Apply Theorem 3.1 with $\Omega = B_1$ and $D = B_{r/R}$. Writing $C_0(K, p) = 2^{2-2/p} C(K, p)$, we obtain

$$\max_{\overline{B_r}} |u| \leq \max_{\partial B_r} |u| + C_0(K, p) R^{2-2/p} \|\tilde{A} : D^2 u\|_{L^p(B_R)}.$$

Since $p > 1$ and $R < 1$, equations (3.3)–(3.4) imply the uniform bound

$$\max_{\overline{B_r}} |u| \leq \max_{\partial B_r} |u| + \frac{C_0(K, p)}{\sqrt{\delta(1-\delta)}} \|A : D^2 u\|_{L^p(B_1)}. \quad (3.5)$$

For each $r < 1$ we may choose $R = (1+r)/2$, so the displayed constant is independent of r .

Finally, let $r \uparrow 1$. Uniform continuity of u on $\overline{B_1}$ and $u(\theta) = 0$ for $|\theta| = 1$ show that

$$\max_{\partial B_r} |u| \leq \sup_{|\theta|=1} |u(r\theta) - u(\theta)| \longrightarrow 0.$$

Also $\max_{\overline{B_r}} |u| \rightarrow \max_{\overline{B_1}} |u|$ by continuity and exhaustion. Passing to the limit in (3.5) proves the desired estimate with a constant depending only on δ and p . All coefficient fields throughout the argument are merely measurable; no coefficient smoothing or boundary regularity of $D^2 u$ was used. $\square$

4 A determinant Mellin estimate

This section proves the following sufficient estimate. As throughout, $a = 1 - (n - 1)\delta$ and $\mathcal{K}_\delta = \{A \in \text{Sym}_n : \text{tr } A = 1, A \geq \delta I\}$.

Theorem 4.1. *Let $n \geq 3$, $0 < \delta < 1/n$, and*

$$\max\{2, n/2, na\} < p < \infty.$$

There is a constant $C = C(n, \delta, p)$ such that, for every measurable $A : B_1 \rightarrow \mathcal{K}_\delta$ and every real $u \in C^2(B_1) \cap C(\overline{B_1})$ with $u = 0$ on ∂B_1 and $A : D^2 u \in L^p(B_1)$,

$$\|u\|_{L^\infty(B_1)} \leq C \|A : D^2 u\|_{L^p(B_1)}. \quad (4.1)$$

The proof first establishes an estimate for positive matrix measures whose double divergence is a finite signed measure, and then applies that estimate to positive functionals associated with a smooth Dirichlet problem. For matrices, $|M|$ denotes the Frobenius norm and $\|M\|_{\text{op}}$ the operator norm. Write

$$\mathcal{C}_\delta = \{M \in \text{Sym}_n : M \geq \delta(\text{tr } M)I\}.$$

Taking traces in this inequality shows that $\text{tr } M \geq 0$; thus $\mathcal{C}_\delta$ is a closed convex cone of positive semidefinite matrices. Its trace-one section is $\mathcal{K}_\delta$. We use $\text{Div}^2 M = \sum_{i,j} \partial_i \partial_j M_{ij}$.

4.1 The shifted determinant inequality

We state explicitly the classical periodic Monge–Ampère solvability result used here. Let $Q = \mathbb{R}^n / \ell \mathbb{Z}^n$ be a flat cubic torus. If S is a constant positive definite matrix and $f \in C^\infty(Q)$ is strictly positive, with

$$\det S = \frac{1}{|Q|} \int_Q f \, dx,$$

then there is a smooth periodic function ϕ , unique after its mean has been fixed, such that

$$S + D^2 \phi > 0, \quad \det(S + D^2 \phi) = f. \quad (4.2)$$

This is the periodic Monge–Ampère theorem of Li [8]; see also the periodic Hessian equation and the proof of Theorem 5.1 in [7]. The determinant compatibility condition above is expressed using normalized torus averages.

Lemma 4.2. *Put $s = 1/(n - 1)$. If $B \in L^\infty(Q; \text{Sym}_n)$ is positive semidefinite almost everywhere and $\text{Div}^2 B = 0$ in distributions, then*

$$\frac{1}{|Q|} \int_Q \det(B)^s \, dx \leq \det(\overline{B})^s, \quad \overline{B} = \frac{1}{|Q|} \int_Q B \, dx. \quad (4.3)$$

In particular, the same inequality holds after replacing B by $B + \lambda I$ for any $\lambda \geq 0$.

Proof. First suppose that B is smooth and positive definite. Set $f = \det(B)^s$ and $J = |Q|^{-1} \int_Q f$. The matrix

$$S = c \overline{B}^{-1}, \quad c = (J \det \overline{B})^{1/n},$$

is positive definite and has determinant J . Apply (4.2) and put $H = S + D^2 \phi$. The arithmetic–geometric mean inequality applied to the positive matrix $B^{1/2} H B^{1/2}$ gives

$$f = (\det B \det H)^{1/n} \leq \frac{1}{n} B : H.$$

Here $(1 + s)/n = s$. Periodic integration by parts twice, using $\text{Div}^2 B = 0$, now gives

$$J \leq \frac{1}{n} \overline{B} : S = c.$$

As $J > 0$, raising to the n th power and dividing by J yields $J^{n-1} \leq \det \overline{B}$, as required. For bounded positive semidefinite B , periodic convolution by a nonnegative approximate identity preserves positivity and the double-divergence condition. Add εI to the convolution, apply the smooth positive definite case, and pass to the limit. The matrices stay bounded and converge almost everywhere along a sequence, so dominated convergence applies to the determinant. Finally, $\text{Div}^2(\lambda I) = 0$. $\square$

Thus the particular case of the Div^2 inequality in [7, Theorem 5.1] needed below follows directly from (4.2), without a stronger first-divergence hypothesis.

4.2 The Mellin integrand and its sign

For the remainder of the matrix argument let

$$s = \frac{1}{n-1}, \quad b = \frac{n}{n-1} = 1 + s, \quad b < q < 2.$$

For a positive semidefinite matrix M , define

$$\mathcal{F}_q(M) = \int_0^\infty \lambda^{q-b-1} \left\{ \det(M + \lambda I)^s - \lambda^b - s(\text{tr } M) \lambda^{b-1} \right\} d\lambda. \quad (4.4)$$

Lemma 4.3. *The integral in (4.4) is absolutely convergent. The function $\mathcal{F}_q$ is continuous on the closed positive semidefinite cone, is homogeneous of degree q , and satisfies*

$$|\mathcal{F}_q(M)| \leq C_{n,q} |M|^q. \quad (4.5)$$

For every $\varepsilon > 0$ there is a finite constant such that

$$|D\mathcal{F}_q(M)| \leq C_{n,q,\varepsilon} |M|^{q-1} \quad \text{if } M > 0, \quad M \geq \varepsilon(\text{tr } M)I. \quad (4.6)$$

Moreover, every bounded positive semidefinite periodic field with $\text{Div}^2 B = 0$ satisfies

$$\int_Q \mathcal{F}_q(B) dx \leq |Q| \mathcal{F}_q(\overline{B}). \quad (4.7)$$

Proof. Uniformly for M in a bounded subset of the positive semidefinite cone, the large- λ expansion is

$$\det(M + \lambda I)^s = \lambda^b + s(\text{tr } M) \lambda^{b-1} + O(\lambda^{b-2}).$$

Consequently the integrand in (4.4) is $O(\lambda^{q-3})$ at infinity, which is integrable because $q < 2$. At zero the determinant term is bounded on bounded matrix sets, and its weight is integrable because $q > b$. The two subtracted terms have powers λ^{q-1} and λ^{q-2} and are also integrable. These uniform bounds give absolute convergence and continuity by dominated convergence. Changing variables $\lambda = t\mu$ proves $\mathcal{F}_q(tM) = t^q \mathcal{F}_q(M)$ for $t > 0$; continuity handles $t = 0$. The growth estimate follows by maximizing on the compact set of positive semidefinite matrices of norm one.

For the derivative estimate, normalize $\text{tr } M = 1$ and restrict to $M \geq \varepsilon I$. On this compact set, differentiation of the bracket in (4.4) is bounded for $0 < \lambda \leq 1$. For large λ , the derivative of the

leading affine term cancels, leaving $O(\lambda^{b-2})$, uniformly on this set. The same integrable bounds justify differentiation under the integral. Homogeneity gives (4.6) at arbitrary nonzero scale.

For each $\lambda > 0$, apply Lemma 4.2 to $B + \lambda I$. The constant and affine trace terms in (4.4) have identical averages on both sides. Subtract them, multiply by $\lambda^{q-b-1} > 0$, and integrate. The absolute bounds already proved, uniform for bounded B , justify Fubini and give (4.7). $\square$

Lemma 4.4. *If*

$$b < q < \min \left\{ 2, \frac{na}{na-1} \right\}, \quad (4.8)$$

then there is $c = c(n, \delta, q) > 0$ such that

$$\mathcal{F}_q(M) \geq c(\operatorname{tr} M)^q \quad (M \in \mathcal{C}_\delta). \quad (4.9)$$

Proof. Consider first $\operatorname{tr} M = 1$. Its eigenvalues are at least δ . If two of them are $\delta + x, \delta + y$, with $x, y > 0$, replacing this pair by $\delta, \delta + x + y$ decreases their shifted product:

$$(\lambda + \delta + x)(\lambda + \delta + y) - (\lambda + \delta)(\lambda + \delta + x + y) = xy > 0.$$

Repeating the operation proves, simultaneously for every $\lambda \geq 0$, that the minimum of $\det(M + \lambda I)$ is attained at $M_* = \operatorname{diag}(a, \delta, \dots, \delta)$. Since the subtracted terms in (4.4) depend only on the trace, $\mathcal{F}_q(M) \geq \mathcal{F}_q(M_*)$.

We compute this last value using absolutely convergent integrals. Write

$$B(\alpha, \beta) = \int_0^\infty t^{\alpha-1} (1+t)^{-\alpha-\beta} dt \quad (\alpha, \beta > 0), \quad c_{n,q} = \frac{b(b-1)}{q(q-1)} B(q-b, 2-q) > 0.$$

The elementary remainder formulas

$$\begin{aligned} (t+1)^b - t^b - bt^{b-1} &= b(b-1) \int_0^1 (1-v)(t+v)^{b-2} dv, \\ (t+1)^{b-1} - t^{b-1} &= (b-1) \int_0^1 (t+v)^{b-2} dv \end{aligned}$$

have nonnegative right sides. Tonelli's theorem and $t = vz$ show that

$$\int_0^\infty t^{q-b-1} (t+v)^{b-2} dt = v^{q-2} B(q-b, 2-q) \quad (v > 0).$$

Integrating in v therefore proves

$$\int_0^\infty t^{q-b-1} [(t+1)^b - t^b - bt^{b-1}] dt = c_{n,q}, \quad (4.10)$$

$$\int_0^\infty t^{q-b-1} [(t+1)^{b-1} - t^{b-1}] dt = \frac{q}{b} c_{n,q}. \quad (4.11)$$

Indeed, the required v integrals are $1/[q(q-1)]$ and $1/(q-1)$.

For M_* ,

$$\det(M_* + \lambda I)^s = (\lambda + a)^s (\lambda + \delta) = (\lambda + a)^b + (\delta - a)(\lambda + a)^{b-1}.$$

The identity $ba + \delta - a = sa + \delta = s$ ensures that the bracket in (4.4) is precisely the sum of the two renormalized brackets in (4.10) and (4.11), after scaling $\lambda = at$. Consequently

$$\mathcal{F}_q(M_*) = c_{n,q} a^{q-1} \left[a + \frac{q}{b} (\delta - a) \right]. \quad (4.12)$$

There is no separation of divergent integrals in this computation. Since $(n-1)(a-\delta) = na-1 > 0$, the final bracket is positive exactly when $q < na/(na-1)$. This proves the positive lower bound on the trace-one section. Homogeneity proves (4.9) for nonzero $M \in \mathcal{C}_\delta$; zero trace in this cone implies $M = 0$. $\square$

4.3 A positive periodic correction

Fix a radius $R > 0$ and a cubic torus Q whose fundamental cube contains $\overline{B_{R+1}}$ in its interior. Its size will remain fixed. A finite scalar signed measure ν has total variation norm denoted by $\|\nu\|_{\mathcal{M}}$.

Lemma 4.5. *For every*

$$1 \leq q < \frac{n}{n-2}$$

there are periodic matrix kernels $R_0, K \in L^q(Q; \text{Sym}_n)$ such that

$$\text{Div}^2 R_0 = \delta_0 - |Q|^{-1}, \quad \text{Div} K = 0, \quad K(x) \geq \|R_0(x)\|_{\text{op}} I \quad \text{for almost every } x. \quad (4.13)$$

For every finite signed measure ν on Q of total mass zero, the fields

$$R = R_0 * \nu, \quad S = K * |\nu|, \quad T = S - R$$

satisfy

$$T \geq 0, \quad \text{Div}^2 T = -\nu, \quad \|T\|_{L^q(Q)} \leq C \|\nu\|_{\mathcal{M}}. \quad (4.14)$$

They also satisfy $\|\bar{T}\|_{\text{op}} \leq C \|\nu\|_{\mathcal{M}}$. If ν has a bounded density, then R, S, T are bounded.

Proof. Let E be the mean-zero periodic fundamental solution of Δ^2 :

$$\Delta^2 E = \delta_0 - |Q|^{-1}.$$

It can be defined by its Fourier coefficients, taking the zero coefficient to be zero and dividing every nonzero Fourier coefficient of δ_0 by the corresponding positive symbol of Δ^2 . Put

$$R_0 = D^2 E, \quad P_0 = (\Delta E)I - D^2 E.$$

Commuting distributional derivatives gives $\text{Div} P_0 = 0$ and $\text{Div}^2 R_0 = \Delta^2 E$.

We verify a positive lower bound for the singular part of $-P_0$. Let $\sigma_{n-1} = |S^{n-1}|$ and $\gamma_n = [(n-2)\sigma_{n-1}]^{-1}$. A Euclidean fundamental solution of Δ^2 is

$$E_{\text{sing}}(x) = \begin{cases} \frac{\gamma_n}{2(n-4)} |x|^{4-n}, & n \neq 4, \\ -\frac{\gamma_n}{2} \log |x|, & n = 4. \end{cases}$$

Indeed, $\Delta E_{\text{sing}} = -\gamma_n |x|^{2-n}$ and $\Delta(-\gamma_n |x|^{2-n}) = \delta_0$ in distributions. Near the origin the periodic E differs from E_{sing} by a smooth function. To see this, subtract a smooth cutoff of E_{sing} equal to it near zero. The double Laplacian of the difference is smooth, and its nonzero Fourier coefficients can be divided by the symbol of Δ^2 ; the resulting coefficients decay faster than every power. Thus the difference is smooth.

Writing $e = x/|x|$, direct differentiation gives the singular matrix parts

$$(R_0)_{\text{sing}}(x) = \frac{\gamma_n}{2} |x|^{2-n} [(n-2)e \otimes e - I], \quad (4.15)$$

$$(-P_0)_{\text{sing}}(x) = \frac{\gamma_n}{2} |x|^{2-n} [I + (n-2)e \otimes e]. \quad (4.16)$$

These formulas also hold in the logarithmic case. The second derivatives displayed are locally integrable and carry no atom at zero: integration by parts across a sphere of radius ε leaves boundary terms of order ε^2 and $O(\varepsilon^3)$ (with a possible logarithm in the latter term), which vanish.

The least eigenvalue of the bracket in (4.16) is one, whereas the operator norm of the bracket in (4.15) is $\max\{1, n-3\}$. Choose $C_1 > \max\{1, n-3\}$. The smooth remainders and compactness away from zero then allow $C_2 > 0$ to be chosen so that

$$K = -C_1 P_0 + C_2 I \geq \|R_0\|_{\text{op}} I$$

everywhere except at the singular point. This is a distributionally Div-free, locally integrable positive kernel. Equations (4.15)–(4.16) show that both kernels belong to $L^q(Q)$ whenever $q(n-2) < n$.

Convolution with finite measures is well defined in these L^q spaces. Since $\nu(Q) = 0$, the first identity in (4.13) gives $\text{Div}^2 R = \nu$. Also $\text{Div} S = 0$. For almost every x and every vector ξ , kernel domination and the total variation inequality give

$$\xi^\top S(x) \xi \geq |\xi|^2 \int_Q \|R_0(x-y)\|_{\text{op}} d|\nu|(y) \geq |\xi^\top R(x) \xi|.$$

This proves $T \geq 0$. Minkowski's integral inequality gives

$$\|R\|_{L^q} \leq \|R_0\|_{L^q} \|\nu\|_{\mathcal{M}}, \quad \|S\|_{L^q} \leq \|K\|_{L^q} \|\nu\|_{\mathcal{M}},$$

and hence (4.14). Distributional periodic derivatives have zero mean, so $\overline{R_0} = \overline{P_0} = 0$ and $\overline{K} = C_2 I$. It follows that $\overline{T} = C_2 \|\nu\|_{\mathcal{M}} I$. Finally, the convolution of an L^1 kernel with a bounded density is bounded. $\square$

4.4 The matrix measure estimate

Lemma 4.6 (Extension from fields to measures). *Let $1 < q < \infty$ and $R > 0$. Suppose that a constant C_0 satisfies*

$$\|\text{tr } M\|_{L^q(\mathbb{R}^n)} \leq C_0 \|\text{Div}^2 M\|_{\mathcal{M}(\mathbb{R}^n)}$$

for every smooth $\mathcal{C}_\delta$ -valued field compactly supported in B_{R+1} . Then every finite symmetric $\mathcal{C}_\delta$ -valued matrix measure M supported in $\overline{B_R}$ with finite signed measure $\text{Div}^2 M$ has $\text{tr } M = g \, dx$ for some $g \in L^q(\mathbb{R}^n)$, and the same estimate holds with $\|g\|_{L^q}$ on the left.

Proof. The measure cone condition means that $M - \delta(\text{tr } M)I$ is a positive semidefinite matrix measure. Taking traces and using $1 - n\delta > 0$ shows that $\text{tr } M$ is nonnegative, and then the cone condition also makes M positive semidefinite. For a nonnegative smooth approximate identity of compact support, $M_\varepsilon = M * \rho_\varepsilon$ is smooth, remains in $\mathcal{C}_\delta$, and is supported in B_{R+1} for small ε . Its double divergence is $(\text{Div}^2 M) * \rho_\varepsilon$, whose total variation is at most $\|\text{Div}^2 M\|_{\mathcal{M}}$. The hypothesized estimate therefore gives a uniform L^q bound on $\text{tr } M_\varepsilon$. Weak compactness in L^q and convergence of mollifications in distributions identify the weak limit as the density of $\text{tr } M$. Weak lower semicontinuity preserves the same bound. $\square$

Proposition 4.7. *Suppose that*

$$\frac{n}{n-1} < q < \min \left\{ 2, \frac{na}{na-1}, \frac{n}{n-2} \right\}. \quad (4.17)$$

For every fixed $R > 0$ there is $C = C(n, \delta, q, R)$ with the following property. Let M be a finite symmetric matrix-valued Radon measure on $\mathbb{R}^n$, supported in $\overline{B_R}$, whose values lie in $\mathcal{C}_\delta$ in the measure sense:

$$M - \delta(\text{tr } M)I \text{ is a positive semidefinite matrix measure.}$$

Assume that $\nu = \text{Div}^2 M$ is a finite signed Radon measure. Then $\text{tr } M = g \, dx$ with $g \in L^q(\mathbb{R}^n)$ and

$$\|g\|_{L^q(\mathbb{R}^n)} \leq C \|\nu\|_{\mathcal{M}(\mathbb{R}^n)}. \quad (4.18)$$

Proof. We begin with a smooth field M compactly supported in the fixed ball. Embed it in the torus used in Lemma 4.5. Then $\nu = \text{Div}^2 M$ is smooth and has total mass zero. Choose a smooth periodic function χ which equals $|x|^2$ on a neighborhood of $\overline{B_R}$. Distributional integration by parts gives

$$2 \int_Q \text{tr } M \, dx = \int_Q \chi \, d\nu \leq \|\chi\|_{L^\infty(Q)} \|\nu\|_{\mathcal{M}}. \quad (4.19)$$

Apply Lemma 4.5 and put $N = M + T$. The fields M, T, N are positive semidefinite, N is bounded, and $\text{Div}^2 N = 0$. Since $|M| \leq \text{tr } M$ for positive semidefinite matrices, (4.19) and the mean estimate for T imply

$$|\overline{N}| \leq C \|\nu\|_{\mathcal{M}}. \quad (4.20)$$

We also need the pointwise perturbation bound

$$|\mathcal{F}_q(M + T) - \mathcal{F}_q(M)| \leq C_{n,\delta,q}(|T||M|^{q-1} + |T|^q), \quad M \in \mathcal{C}_\delta, \, T \geq 0. \quad (4.21)$$

If $M \neq 0$ and $|T| \leq |M|$, then, for $0 \leq t \leq 1$,

$$M + tT \geq \delta(\text{tr } M)I, \quad \text{tr}(M + tT) \leq (1 + \sqrt{n}) \text{tr } M, \quad |M + tT| \leq 2|M|.$$

The segment therefore lies in a fixed strict positive cone, and (4.6) and the fundamental theorem of calculus give the first term in (4.21). If $|T| > |M|$, the growth bound (4.5) gives a bound by $C|T|^q$. The case $M = 0$ follows from the same growth bound.

Set $X = \|\text{tr } M\|_{L^q(Q)}$ and $V = \|\nu\|_{\mathcal{M}}$. Lemma 4.4, the perturbation estimate, Hölder's inequality, and Lemma 4.3 give

$$\begin{aligned} cX^q &\leq \int_Q \mathcal{F}_q(M) \, dx \\ &\leq \int_Q \mathcal{F}_q(N) \, dx + C\|T\|_{L^q} X^{q-1} + C\|T\|_{L^q}^q \\ &\leq |Q|\mathcal{F}_q(\overline{N}) + C\|T\|_{L^q} X^{q-1} + C\|T\|_{L^q}^q \\ &\leq CV^q + CVX^{q-1}. \end{aligned}$$

The final line follows from (4.5), (4.20), and (4.14). Young's inequality absorbs half of cX^q and proves (4.18) for smooth M . All constants depend only on the displayed parameters and the fixed support ball.

Apply the smooth estimate with the fixed larger support ball B_{R+1} . Lemma 4.6 then proves (4.18) for all the measures in the statement. No absolute continuity assumption on the original matrix measure is needed. $\square$

4.5 Positive functionals and the Dirichlet estimate

We use one classical linear elliptic solvability fact. On a smooth bounded ball, a smooth uniformly elliptic coefficient field and smooth right-hand side and boundary data admit a unique smooth solution of the Dirichlet problem for $A : D^2$, smooth up to the boundary. Such solutions satisfy the classical maximum and minimum principles. This follows from the Schauder Dirichlet theory and regularity; see [5, Chapter 6]. Only existence, uniqueness, and positivity are used here. No bound involving coefficient derivatives enters the argument.

Proposition 4.8 (Transfer of a matrix measure estimate). *Let $1 < q < \infty$ and $p = q/(q - 1)$. Suppose that every finite symmetric $\mathcal{C}_\delta$ -valued measure M supported in $\overline{B_1}$ with finite signed measure $\text{Div}^2 M$ has trace $g \, dx$ satisfying*

$$\|g\|_{L^q(\mathbb{R}^n)} \leq C_0 \|\text{Div}^2 M\|_{\mathcal{M}(\mathbb{R}^n)}. \quad (4.22)$$

Then every measurable $A : B_1 \rightarrow \mathcal{K}_\delta$ and every real $u \in C^2(B_1) \cap C(\overline{B_1})$ with zero boundary values and $A : D^2 u \in L^p(B_1)$ satisfy

$$\|u\|_{L^\infty(B_1)} \leq 2C_0 \|A : D^2 u\|_{L^p(B_1)}.$$

Proof. We first assume that A is smooth on a neighborhood of $\overline{B_r}$, where $0 < r < 1$. Fix $x \in B_r$ and write $\Omega = B_r$. For smooth real f on $\overline{\Omega}$, let v_f be the classical solution of

$$-A : D^2 v_f = f \quad \text{in } \Omega, \quad v_f = 0 \quad \text{on } \partial\Omega,$$

and set $T_x(f) = v_f(x)$. This is a positive linear functional: if $f \geq 0$, the minimum principle gives $v_f \geq 0$. The trace normalization supplies the exact barrier

$$b_r(y) = \frac{r^2 - |y|^2}{2}, \quad -A : D^2 b_r = 1.$$

Comparison therefore gives $|T_x(f)| \leq b_r(x) \|f\|_{L^\infty(\overline{\Omega})}$. Smooth functions are uniformly dense in $C(\overline{\Omega})$, so T_x extends to a bounded positive functional there. The Riesz–Markov representation theorem provides a finite positive measure μ_x on $\overline{\Omega}$ such that

$$T_x(f) = \int_{\overline{\Omega}} f \, d\mu_x, \quad \mu_x(\overline{\Omega}) = b_r(x).$$

The extension is positive, for example by uniformly approximating each nonnegative continuous function by smooth functions with an added error bound.

Similarly, evaluating at x the solution of $A : D^2 h = 0$ with smooth boundary data defines a positive functional H_x on the boundary data. The maximum principle extends it to $C(\partial\Omega)$ with norm one, and $H_x(1) = 1$. Thus there is a probability measure ω_x on $\partial\Omega$ with $H_x(\psi) = \int_{\partial\Omega} \psi \, d\omega_x$.

For a smooth function ϕ on a neighborhood of $\overline{\Omega}$, decompose it into its harmonic extension from the boundary and a function with zero boundary values. Uniqueness of the Dirichlet solution gives

$$\int_{\overline{\Omega}} A : D^2 \phi \, d\mu_x = \int_{\partial\Omega} \phi \, d\omega_x - \phi(x). \quad (4.23)$$

Define the symmetric matrix measure $M_x = A\mu_x$, extended by zero to $\mathbb{R}^n$. It has trace μ_x , is supported in $\overline{B_r}$, and lies in $\mathcal{C}_\delta$ in the measure sense. Identity (4.23) says exactly that

$$\text{Div}^2 M_x = \omega_x - \delta_x.$$

This signed measure has total variation at most two. Hypothesis (4.22), with the one fixed support ball $\overline{B_1}$, now gives

$$\mu_x = g_x \, dy, \quad \|g_x\|_{L^q(\mathbb{R}^n)} \leq 2C_0, \quad (4.24)$$

uniformly in r, x, A . The conclusion itself excludes a boundary part of μ_x ; no assertion about a Green density was used to construct the measure.

For any u of class C^2 on a neighborhood of $\bar{\Omega}$, approximation in $C^2(\bar{\Omega})$ extends (4.23) to u . Equivalently,

$$u(x) = \int_{\partial\Omega} u \, d\omega_x - \int_{\Omega} A : D^2 u(y) g_x(y) \, dy.$$

The first measure is a probability and the second density satisfies (4.24). Hölder's inequality followed by the supremum over x proves

$$\|u\|_{L^\infty(B_r)} \leq \|u\|_{L^\infty(\partial B_r)} + 2C_0 \|A : D^2 u\|_{L^p(B_r)}. \quad (4.25)$$

The bound is independent of r and of the derivatives of A .

Extend the measurable coefficient field A to $\mathbb{R}^n$ by I/n outside B_1 , changing it on a null set if needed. Convolution with nonnegative smooth approximate identities gives smooth fields A_j with values in the closed convex set $\mathcal{K}_\delta$, converging to A almost everywhere in B_1 . Fix $0 < r < 1$. The given u is of class C^2 on a neighborhood of $\bar{B}_r$, so the smooth-coefficient estimate (4.25) applies to A_j, u there. Since $D^2 u$ is bounded on $\bar{B}_r$ and the coefficient fields are uniformly bounded, dominated convergence gives

$$A_j : D^2 u \longrightarrow A : D^2 u \quad \text{in } L^p(B_r).$$

Letting $j \rightarrow \infty$ in (4.25) proves the same inequality for the original measurable field on B_r , with a constant independent of r .

Now let $r \uparrow 1$. Continuity of u on $\bar{B}_1$ and its zero boundary values imply $\|u\|_{L^\infty(\partial B_r)} \rightarrow 0$. The supremum on B_r increases to that on B_1 , and the L^p source norms tend to the norm on B_1 . This proves the asserted bound. In particular, this approximation only uses boundedness of $D^2 u$ on interior balls and does not impose a global $W^{2,p}$ hypothesis. $\square$

Proof of Theorem 4.1. First suppose that

$$\max\{2, n/2, na\} < p < n.$$

For $q = p/(p-1)$, these inequalities are equivalent to (4.17): the respective upper bounds on q correspond to $p > 2$, $p > na$, and $p > n/2$, while its lower bound corresponds to $p < n$. Proposition 4.7 supplies the hypothesis of Proposition 4.8, so (4.1) follows.

Finally, $\max\{2, n/2, na\} < n$ because $n \geq 3$ and $0 < \delta < 1/n$. Choose

$$p_0 \in (\max\{2, n/2, na\}, n).$$

The already proved estimate at p_0 , followed by Hölder's inequality on B_1 , gives (4.1) for every $p \geq n$ as well. $\square$

Corollary 4.9 (The higher-dimensional threshold). *For every $n \geq 4$, every $0 < \delta < 1/n$ and every $1 \leq p < \infty$, (1.2) holds.*

Proof. Necessity, including the endpoint, is Theorem 2.1. Since $n/2 \geq 2$, the sufficient range in Theorem 4.1 is exactly $p > P(n, \delta)$. $\square$

5 Transfer from a homogeneous Hessian inequality

The following result isolates the analytic step used below. Its hypothesis concerns one integrand for the entire coefficient class. Throughout this section, the matrix norm is the Frobenius norm and $A : H = \text{tr}(AH)$.

Theorem 5.1 (Transfer principle). *Let $n \geq 3$, $0 < \delta < 1/n$, and $n/2 < p < \infty$, and write*

$$\mathcal{K}_\delta = \{A \in \text{Sym}_n : A \geq \delta I, \text{tr } A = 1\}.$$

Suppose that $F : \text{Sym}_n \rightarrow \mathbb{R}$ is continuous, satisfies $F(tH) = t^p F(H)$ for $t \geq 0$, and has the following two properties:

$$\int_{\mathbb{R}^n} F(D^2 \varphi) \, dx \geq 0 \quad \text{for every real } \varphi \in C_c^\infty(\mathbb{R}^n), \quad (5.1)$$

$$F(H) < 0 \quad \text{if } H \neq 0 \text{ and } A : H = 0 \text{ for some } A \in \mathcal{K}_\delta. \quad (5.2)$$

Then there is a finite constant $C = C(n, p, \delta, F)$ such that

$$\max_{\overline{B_1}} |u| \leq C \|A : D^2 u\|_{L^p(B_1)} \quad (5.3)$$

for every measurable $A : B_1 \rightarrow \mathcal{K}_\delta$ and every real $u \in C^2(B_1) \cap C(\overline{B_1})$ satisfying $u = 0$ on ∂B_1 and $A : D^2 u \in L^p(B_1)$. If the choice of F is fixed by n, p , the constant depends only on n, p, δ .

Proof. We first obtain a uniform algebraic estimate. The function

$$h_\delta(H) = \min_{A \in \mathcal{K}_\delta} |A : H|$$

is continuous. The set $Z = \{H : \|H\|_F = 1, h_\delta(H) = 0\}$ is compact and nonempty; for example, it contains every unit trace-free matrix, since $I/n \in \mathcal{K}_\delta$. By (5.2), $b = -\max_Z F > 0$. Compactness and continuity give $\varepsilon > 0$ such that

$$\|H\|_F = 1, \quad h_\delta(H) \leq \varepsilon \quad \implies \quad F(H) \leq -b/2.$$

Put $c = b/2$ and $M = \max_{\|H\|_F=1} |F(H)|$. For each $A \in \mathcal{K}_\delta$, separating the cases $|A : H| \leq \varepsilon$ and $|A : H| > \varepsilon$ on the unit sphere, and then using homogeneity, gives

$$F(H) \leq C_1 |A : H|^p - c \|H\|_F^p, \quad C_1 = (M + c)\varepsilon^{-p}. \quad (5.4)$$

In particular, the constants in this inequality do not depend on A .

Extension to Newton Hessians. Let $T = D^2 \Delta^{-1}$ on real scalar $L^p(\mathbb{R}^n)$, with Fourier symbol

$$\widehat{T_{ij}h}(\xi) = \frac{\xi_i \xi_j}{|\xi|^2} \widehat{h}(\xi).$$

The classical singular integral estimates show that $T : L^p(\mathbb{R}^n) \rightarrow L^p(\mathbb{R}^n; \text{Sym}_n)$ is bounded for $1 < p < \infty$. Moreover,

$$\text{tr}(Th) = h. \quad (5.5)$$

We claim that (5.1) implies

$$\int_{\mathbb{R}^n} F(Th) \, dx \geq 0 \quad (h \in L^p(\mathbb{R}^n; \mathbb{R})). \quad (5.6)$$

First let $h \in C_c^\infty(\mathbb{R}^n)$ and set $v = \Phi * h$, where

$$\Phi(x) = -\frac{|x|^{2-n}}{(n-2)|S^{n-1}|}, \quad \Delta \Phi = \delta_0.$$

Then v is smooth and $D^2v = Th$. For large $|x|$, differentiation under the convolution gives

$$|v(x)| \leq C_h |x|^{2-n}, \quad |Dv(x)| \leq C_h |x|^{1-n}, \quad |D^2v(x)| \leq C_h |x|^{-n}.$$

Choose $\chi \in C_c^\infty(B_2)$ equal to one on B_1 , and put $\varphi_R = \chi(x/R)v$. The term $(\chi(x/R) - 1)D^2v$ tends to zero in L^p . Each remaining term in $D^2\varphi_R - D^2v$ is supported in $B_{2R} \setminus B_R$ and bounded by $C_h R^{-n}$; its L^p norm therefore tends to zero like $R^{n/p-n}$. Consequently

$$D^2\varphi_R \longrightarrow Th \quad \text{in } L^p. \quad (5.7)$$

This argument requires only $p > 1$ and imposes no zero-integral condition on h .

Continuity and homogeneity imply $|F(H)| \leq M\|H\|_F^p$. They also imply, for each $\eta > 0$, an inequality

$$|F(X) - F(Y)| \leq \eta(\|X\|_F^p + \|Y\|_F^p) + C_\eta\|X - Y\|_F^p. \quad (5.8)$$

Indeed, when $\|X - Y\|_F$ is small relative to $\max(\|X\|_F, \|Y\|_F)$, rescaling reduces the assertion to uniform continuity on the unit ball. In the complementary case the growth bound gives the second term. Integrating (5.8), first passing to the limit under strong L^p convergence and then letting $\eta \downarrow 0$, shows that F induces a continuous map from L^p to L^1 . Thus (5.7) proves (5.6) for $h \in C_c^\infty$. Density of real C_c^∞ functions in scalar L^p , boundedness of T , and the same continuity argument prove the claim for arbitrary $h \in L^p$.

A whole-space inverse. For a measurable field $A : \mathbb{R}^n \rightarrow \mathcal{K}_\delta$, define the bounded linear operator on the real Banach space $X = L^p(\mathbb{R}^n; \mathbb{R})$ by

$$P_A h = A : Th.$$

Integrating (5.4) and using (5.6) yields $c\|Th\|_{L^p}^p \leq C_1\|P_A h\|_{L^p}^p$. Since $|\operatorname{tr} H| \leq \sqrt{n}\|H\|_F$, (5.5) gives

$$\|h\|_{L^p} \leq C_0\|P_A h\|_{L^p}, \quad C_0 = \sqrt{n}(C_1/c)^{1/p}. \quad (5.9)$$

This estimate is uniform over all measurable fields in the coefficient class.

To prove surjectivity, set $A_t = (1 - t)I/n + tA$ and $P_t = P_{A_t}$. The convexity of $\mathcal{K}_\delta$ ensures that (5.9) holds for every $t \in [0, 1]$. Also $P_0 h = h/n$, and $\|P_t - P_s\|_{X \rightarrow X} \leq D|t - s|$, where D is bounded solely in terms of the coefficient class and $\|T\|_{L^p \rightarrow L^p}$. Whenever P_t is invertible, its inverse has norm at most C_0 . Choose an integer $N > C_0 D$. The factorization

$$P_{(k+1)/N} = [I + (P_{(k+1)/N} - P_{k/N})P_{k/N}^{-1}]P_{k/N}$$

and a Neumann series prove invertibility successively at k/N , $k = 0, \dots, N$. Hence P_A is bijective and

$$A : Th = f, \quad \|h\|_{L^p} \leq C_0\|f\|_{L^p} \quad (5.10)$$

has a unique scalar solution $h \in L^p$ for every real $f \in L^p$. This continuation uses neither a maximum principle nor a quotient by affine functions.

Positive barriers for smooth coefficients. Suppose now that A is smooth on $\mathbb{R}^n$ and equals I/n outside B_2 . For $0 \leq g \in C_c^\infty(B_2)$, solve (5.10) with $f = -g$. Outside B_2 , the equation and (5.5) give $h/n = 0$, so h is supported in $\overline{B_2}$. Its Newton potential $v = \Phi * h$ satisfies

$$v \in C_0(\mathbb{R}^n), \quad \|v\|_{L^\infty} \leq C_{n,p}\|h\|_{L^p} \leq C_{n,p}C_0\|g\|_{L^p}. \quad (5.11)$$

Here $C_0(\mathbb{R}^n)$ denotes continuous functions vanishing at infinity. For clarity, the estimate uses only the finite quantity

$$\sup_{x \in \mathbb{R}^n} \left(\int_{B_2} |x - y|^{-(n-2)p'} dy \right)^{1/p'}, \quad p' = \frac{p}{p-1}.$$

It is finite because $(n-2)p' < n$, which is equivalent to $p > n/2$: inside a fixed ball, translate and enlarge the integration region to a fixed ball about the singularity; outside that ball, use distance from B_2 . Truncating the singularity and applying the same local bound proves continuity. Compact support of h gives decay at infinity.

In distributions, $D^2v = Th$, so $v \in W_{\text{loc}}^{2,p}$ and $A : D^2v = -g$. Interior regularity for smooth elliptic operators makes v smooth. The classical minimum principle on expanding balls, followed by the decay at infinity, gives $v \geq 0$. Thus (5.11) supplies a positive barrier with a bound independent of derivatives of A . Smooth regularity is used only to justify classical comparison, not to estimate its size.

Passage to the stated coefficient and function classes. Extend the original field A by I/n outside B_1 and mollify it with nonnegative approximate identities of radii tending to zero. The resulting fields A_j are smooth, belong to $\mathcal{K}_\delta$, equal I/n outside B_2 for all sufficiently large j , and converge to A almost everywhere. Fix $0 < R < 1$ and set $f_j = A_j : D^2u$ on $\overline{B_R}$. These functions are continuous. Since D^2u is bounded on this compact ball, dominated convergence gives

$$f_j \longrightarrow A : D^2u \quad \text{in } L^p(B_R). \quad (5.12)$$

For fixed R, j , there exist $0 \leq g_m \in C_c^\infty(B_2)$ such that

$$\epsilon_m := \sup_{\overline{B_R}} |g_m - |f_j|| \longrightarrow 0, \quad \|g_m\|_{L^p(\mathbb{R}^n)} \longrightarrow \|f_j\|_{L^p(B_R)}.$$

To see this, retain $|f_j|$ on $\overline{B_R}$, extend its boundary values radially through a shrinking exterior shell, and multiply by a nonnegative continuous cutoff which vanishes at the outer edge of that shell. These continuous extensions have uniformly bounded height and vanishing exterior L^p mass. A subsequent nonnegative mollification, chosen sufficiently fine for each extension, gives the asserted uniform and norm approximations.

Let v_m be the positive barrier for A_j and g_m , and put

$$b_R = \max_{\partial B_R} |u|, \quad q_R(x) = \frac{R^2 - |x|^2}{2}.$$

The trace normalization gives $A_j : D^2q_R = -1$. Therefore

$$A_j : D^2(\pm u - v_m - b_R - \epsilon_m q_R) = \pm f_j + g_m + \epsilon_m \geq 0.$$

The compared functions are nonpositive on ∂B_R , since $v_m \geq 0$. Classical comparison on B_R consequently gives

$$\max_{\overline{B_R}} |u| \leq b_R + C_{n,p} C_0 \|g_m\|_{L^p(\mathbb{R}^n)} + \epsilon_m R^2/2.$$

Let first $m \rightarrow \infty$, then $j \rightarrow \infty$ using (5.12), and finally $R \uparrow 1$. Continuity on $\overline{B_1}$ and the zero boundary values imply $b_R \rightarrow 0$. This proves (5.3). All constants are independent of A, u, R, j, m ; in particular, no global Sobolev regularity or boundary differentiability of the original u has been required. $\square$

6 A planar integral inequality and its slices

The following input will be used in the last section. For $1 < p < 2$ and $K \in \text{Sym}_2$, let $M = \|K\|_{\text{op}}$, $D = \det K$, and define

$$B_p(K) = \frac{(2-p)M^p + pDM^{p-2}}{2(p-1)}, \quad B_p(0) = 0. \quad (6.1)$$

This is a continuous, even, orthogonally invariant function, homogeneous of degree p . Indeed, $|D| \leq M^2$ gives continuity at zero. If $t = |\text{tr } K|/2$ and r is half the absolute eigenvalue gap, then $M = t + r$ and $D = t^2 - r^2$, so equivalently

$$B_p(K) = \left(\frac{t}{p-1} - r \right) (t+r)^{p-1}.$$

Theorem 6.1 (Planar Burkholder integral inequality). *For $1 < p < 2$ and every real $\varphi \in C_c^\infty(\mathbb{R}^2)$,*

$$\int_{\mathbb{R}^2} B_p(D^2\varphi) \, dx \geq 0. \quad (6.2)$$

This is the zero-background case of Guerra's Hessian quasiconvexity theorem [6, Corollary 1.3]. The normalization in (6.1) agrees with the equivalent trace-gap formula above. Only (6.2), rather than any stronger higher-dimensional assertion, is imported from that theorem.

Lemma 6.2 (Slicing). *Let $n \geq 2$, and average over the two-dimensional linear subspaces $E \subset \mathbb{R}^n$ with their rotation-invariant probability measure. Set*

$$\mathcal{B}_{n,p}(H) = \int_{\text{Gr}(2,n)} B_p(H|_E) \, dE.$$

Then $\mathcal{B}_{n,p}$ is continuous, even, orthogonally invariant and p -homogeneous, and

$$\int_{\mathbb{R}^n} \mathcal{B}_{n,p}(D^2\varphi) \, dx \geq 0 \quad (\varphi \in C_c^\infty(\mathbb{R}^n; \mathbb{R})).$$

Proof. Fix E and decompose $x = y + z$ with $y \in E$, $z \in E^\perp$. For each z , the restriction $y \mapsto \varphi(y+z)$ is a smooth compactly supported planar function with Hessian $(D^2\varphi)|_E$. Apply Theorem 6.1 and integrate over z . The bound $|B_p(H|_E)| \leq C_p|H|^p$ justifies Fubini and then integration over E . The remaining properties follow directly from the definition. $\square$

7 The remaining dimension: three-dimensional results

This section gives computer-assisted extensions of the sufficient range in dimension three. The finite certificates are part of the accompanying supplement. We give the analytic bounds, the exact verification algorithms, the domains of every certificate, and the resulting conclusions. In particular, the numerical search that preceded the certificates is not an input to the proofs.

Theorem 7.1 (Three-dimensional ranges). *Retain all coefficient, boundary, regularity, and integrability hypotheses of the problem, with $n = 3$.*

1. *If*

$$0 < \delta \leq \frac{9}{40} \quad \text{or} \quad \frac{128}{499} \leq \delta < \frac{1}{3},$$

the asserted estimate holds if and only if $p > \max\{3/2, 3(1-2\delta)\}$.

2. *For every $9/40 < \delta < 1/3$, the estimate also holds for every $p \geq 33/20$.*

The portion $\delta \leq 1/6$ follows from Theorems 4.1 and 2.1. The remaining sufficient assertions are computer-assisted, with the exact finite certificates described below.

7.1 The averaged planar integrand and its null cone

For $1 < p < 2$, let B_p be the planar integrand of Theorem 6.1, in the normalization

$$B_p(K) = \frac{(2-p)M^p + pDM^{p-2}}{2(p-1)}, \quad M = \|K\|_{\text{op}}, \quad D = \det K, \quad B_p(0) = 0. \quad (7.1)$$

For $H \in \text{Sym}_3$, define

$$\Phi_p(H) = \frac{1}{4\pi} \int_{S^2} B_p(H|_{\nu^\perp}) d\nu. \quad (7.2)$$

This function is continuous, even, orthogonally invariant, and homogeneous of degree p . Lemma 6.2 gives

$$\int_{\mathbb{R}^3} \Phi_p(D^2 v) dx \geq 0 \quad (v \in C_c^\infty(\mathbb{R}^3; \mathbb{R})). \quad (7.3)$$

Thus, for $p > 3/2$, Theorem 5.1 applies as soon as Φ_p is strictly negative on every nonzero matrix in the coefficient null cone.

Put

$$p_0 = 3(1 - 2\delta), \quad S = \frac{2p_0}{3 - p_0} = \frac{1}{\delta} - 2, \quad T = S - 1. \quad (7.4)$$

By Lemma 1.2, the null-cone condition is

$$\delta \text{tr } H + (1 - 3\delta)\lambda_{\min} H \leq 0 \leq \delta \text{tr } H + (1 - 3\delta)\lambda_{\max} H. \quad (7.5)$$

After replacing H by $-H$, assume $\text{tr } H \geq 0$. If $H \neq 0$, its least eigenvalue is strictly negative. Scaling and rotation reduce (7.5) exactly to the compact set

$$H = \text{diag}(-1, b, c), \quad -1 \leq b \leq c, \quad 1 \leq b + c \leq S. \quad (7.6)$$

The upper inequality in (7.5) is then automatic. For fixed p_0 , strict negativity on a compact part of this set at exponent p_0 persists at all sufficiently close exponents.

Two elementary identities will be used repeatedly. For fixed D ,

$$\frac{\partial}{\partial M} B_p(M, D) = \frac{p(2-p)}{2(p-1)} M^{p-3} (M^2 - D) \geq 0 \quad \text{if } M^2 \geq |D|. \quad (7.7)$$

Also, if $-v \leq u \leq v$ and $v > 0$, compression of $\text{diag}(u, v, v)$ has eigenvalues v and $u + (v - u)\nu_1^2$. Since $\mathbb{E}\nu_1^2 = 1/3$,

$$\Phi_p(\text{diag}(u, v, v)) = \frac{[pu + (3-p)v]v^{p-1}}{3(p-1)}. \quad (7.8)$$

Here and below $\mathbb{E}$ denotes normalized angular averaging.

7.2 The sector with $b \geq 1$

Lemma 7.2. *Suppose $33/20 \leq p_0 < 2$, $p_0 < p < 2$, and $1 \leq b \leq c$, $b + c \leq 2p_0/(3 - p_0)$. Then*

$$\Phi_p(\text{diag}(-1, b, c)) \leq \Phi_p(\text{diag}(-1, m, m)) < 0, \quad m = (b + c)/2.$$

The first inequality also holds at $p = p_0$; at that exponent the right side vanishes only if $m = p_0/(3 - p_0)$.

Proof. For a planar matrix of positive trace with eigenvalues $\lambda \leq \mu$, put $r = (\mu - \lambda)/2$. Differentiating (7.1), in an eigenbasis and then by continuity at a repeated eigenvalue, gives

$$D^2 B_p(K)[Q, Q] = p\mu^{p-3}\{\mu \det Q + (2-p)r q_{22}^2\}. \quad (7.9)$$

Take $D_0 = \text{diag}(0, 1, -1)$. Its compression satisfies $\det(D_0|_{\nu^\perp}) = -\nu_1^2$ and $|q_{22}| \leq 1$. If $b > 1$, every compression of H has positive trace, $b \leq \mu \leq c$, and $\lambda \geq -1$. Consequently

$$D^2 \Phi_p(H)[D_0, D_0] \leq p\{(2-p)b^{p-2} - c^{p-2}/3\}. \quad (7.10)$$

For $p_0 \geq 9/5$, $c/b \leq T < 3$, so the right side is strictly negative: $3(2-p)(c/b)^{2-p} \leq (3/5)3^{1/5} < 1$.

For the smaller parameters use $H \geq bI - (b+1)e_1 e_1^T$, which yields the stronger compression bound

$$\lambda \geq -1 + (b+1)\nu_1^2.$$

Writing $q = 2 - p$ in (7.9), and bounding separately the positive and negative terms, gives

$$\begin{aligned} D^2 \Phi_p(H)[D_0, D_0] &\leq pc^{-q} \left\{ -\frac{1}{3} - \frac{q(b+1)}{6c} + q(c/b)^q \frac{b+1}{2b} \right\} \\ &\leq \frac{pc^{-q}}{3} \{q(3T^q - 1/T) - 1\}, \quad T = b + c - 1. \end{aligned} \quad (7.11)$$

For example, the negative terms use $\mu^{-q} \geq c^{-q}$ and $\mu^{-q-1} \geq c^{-q-1}$, while the positive terms use the corresponding upper bounds at b . The last step uses $c/b \leq T$, $(b+1)/(2b) \leq 1$, and $(b+1)/(2c) \geq 1/T$. The function $q(3T^q - 1/T)$ increases in both variables for $q > 0, T \geq 1$. The following rational bounds therefore prove strict concavity throughout the remaining parameter bands:

p_0 band	upper bound for q	upper bound for T	upper bound for $q(3T^q - 1/T)$
$[17/10, 9/5]$	$3/10$	2	$39/40$
$[5/3, 17/10]$	$1/3$	$21/13$	$313/315$
$[33/20, 5/3]$	$7/20$	$3/2$	$119/120$

The bounds are strict and follow respectively from $2^{3/10} < 5/4$, $(21/13)^{1/3} < 6/5$, and $(3/2)^{7/20} < 7/6$. These power comparisons are the integer inequalities $2^3 4^{10} < 5^{10}$, $21 \cdot 125 < 13 \cdot 216$, and $3^7 6^{20} < 2^7 7^{20}$.

At fixed $b + c$, the function of the splitting parameter is even and strictly concave. It is therefore maximized at $b = c = m$. The assertion for $b = 1$ follows by continuity. Finally $m \leq p_0/(3 - p_0) < p/(3 - p)$ and (7.8) give the required sign. The same concavity bounds hold at $p = p_0$, and (7.8) gives the stated equality condition. $\square$

7.3 Compression estimates for the sector with $b \leq 1$

We next prove the following finite-certificate assertion.

Proposition 7.3 (Computer-assisted cone sign). *For $33/20 \leq p_0 < 2$, every matrix in (7.6) with $b \leq 1$ satisfies $\Phi_{p_0}(H) < 0$.*

The proof occupies this and the next two subsections. All uses of finite verification are isolated in Tables 1 and 2.

First, $C = \|H\|_{\text{op}}$ and (7.7) give

$$\Phi_p(H) \leq \frac{C^p}{2(p-1)} \left\{ 2 - p + \frac{p\sigma_2(H)}{3C^2} \right\}, \quad \sigma_2(H) = bc - b - c, \quad (7.12)$$

because $\mathbb{E} \det(H|_{\nu^\perp}) = \sigma_2(H)/3$. If $c \leq 1$, then $C = 1$ and $b \geq 1 - c$ imply $-\sigma_2 \geq c^2 - c + 1 \geq 3/4$. The bound is strictly negative for $p > 8/5$, covering every parameter used below.

When $p_0 \geq 9/5$ and $b \leq 0$, write $x = -b \in [0, 1]$. The ratio $-\sigma_2/c^2$ decreases in $c \geq 1 - b$, and hence is at least

$$\frac{x^2 + Sx + S}{(x + S)^2} \geq \frac{1}{S}.$$

The last inequality follows by differentiation in x , since $S \geq 3$. At $p = p_0$, the braces in (7.12) are at most $(9 - 5p_0)/6 \leq 0$. Equality is possible only at $p_0 = 9/5, b = 0, c = 3$. Almost every plane compression of that matrix has norm strictly less than 3, making (7.12) strict.

There is also an analytic upper-parameter range. If $T \geq 2$ and $b \leq 1$, then

$$-\sigma_2(H)/C^2 \geq 1/T^2. \quad (7.13)$$

For $c \leq 1$ this follows from the preceding $3/4$ bound. For $1 \leq c \leq T$, use $\sigma_2 \leq -1$. For $c \geq T$, use $b \leq T + 1 - c$ and

$$-\sigma_2/c^2 \geq 1 - (T + 1)/c + (T + 1)/c^2,$$

whose derivative is nonnegative for $c \geq T \geq 2$. Equality in (7.13) is possible only at $b = 1, c = T$. Therefore the critical sign holds if $p_0 \geq 6T^2/(3T^2 + 1)$, including equality by the same strict compression-norm argument. Substitution of (7.4) reduces this condition to

$$p_0 \geq \rho := \frac{9 + 3\sqrt{2}}{7} < \frac{19}{10}. \quad (7.14)$$

For the remaining sectors condition on $z = |\nu_3|$, uniform on $[0, 1]$. Define

$$m = c - (c - b)z^2, \quad M_0 = \max(1, m), \quad \overline{D} = \frac{c(b - 1)}{2}(1 - z^2) - bz^2. \quad (7.15)$$

The comparison $H \leq \text{diag}(b, b, c)$ bounds the larger compression eigenvalue by m ; the smaller eigenvalue is at least -1 . Thus the actual compressed norm is at most M_0 . Applying (7.7) before the conditional angular average gives

$$2(p - 1)\Phi_p(H) \leq (2 - p) \int_0^1 M_0^p dz + p \int_0^1 \overline{D} M_0^{p-2} dz. \quad (7.16)$$

Write $d = c - 1$, $k = c - b$, $t = \sqrt{d/k}$, and $\Delta_0 = c - M_0 = \min(d, kz^2)$. For $c \geq 1, b \leq 1$,

$$M_j := \int_0^1 \Delta_0^j dz = d^j \left(1 - \frac{2j}{2j+1}t\right), \quad j \geq 1. \quad (7.17)$$

At $c = 1$ these moments are zero, including $b = c = 1$. Taylor's theorem, since the second derivative of r^p is at most $p(p - 1)$ on $r \geq 1$, gives

$$\int_0^1 M_0^p dz \leq c^p - pc^{p-1}M_1 + \frac{p(p-1)}{2}M_2. \quad (7.18)$$

7.4 The nonnegative-middle sector and its five certificates

Assume $0 \leq b \leq 1 \leq c$. Then $\overline{D} \leq 0$. For $9/5 \leq p \leq 19/10$, discard part of this negative term using $M_0^{p-2} \geq c^{p-2}$ in (7.16). Equations (7.17)–(7.18) yield

$$\frac{2(p-1)}{c^p} \Phi_p(H) \leq (2-p) - \frac{p(2-p)}{c} M_1 + \frac{p(p-1)(2-p)}{2c^{9/5}} M_2 + \frac{p(bc - b - c)}{3c^2}. \quad (7.19)$$

The replacement $c^p \geq c^{9/5}$ was made only in a positive term.

For smaller exponents retain the determinant weight. Put $q = 2 - p$ and $X = 1 - M_0/c$. Exact moments, obtained by splitting at $z = t$, are

$$\begin{aligned} R_j &:= \int_0^1 X^j dz = (d/c)^j \left(1 - \frac{2j}{2j+1}t\right), \\ H_j &:= \int_0^1 z^2 X^j dz = (d/c)^j \left(\frac{1}{3} - \frac{2j}{3(2j+3)}t^3\right), \\ S_j &:= \int_0^1 \frac{-\overline{D}}{c^2} X^j dz = \frac{1-b}{2c}(R_j - H_j) + \frac{b}{c^2}H_j \geq 0. \end{aligned} \quad (7.20)$$

Here $R_0 = 1, H_0 = 1/3, S_0 = -\sigma_2/(3c^2)$. Define positive coefficients by

$$a_2(p) = p(p-1)/2, \quad a_{j+1} = a_j(j-p)/(j+1), \quad d_1(q) = q, \quad d_{j+1} = d_j(q+j)/(j+1).$$

The binomial series for $(1-X)^p$ has constant and linear terms $1 - pX$ and positive coefficients a_j thereafter. Its tail after order N is at most $a_{N+1}X^{N+1}/(1-X) \leq a_{N+1}cX^{N+1}$. The series for $(1-X)^{-q}$ has positive coefficients d_j and can be truncated from below in the negative determinant term. Hence

$$\frac{2(p-1)}{c^p} \Phi_p(H) \leq U_{p,N}, \quad (7.21)$$

where

$$U_{p,N} = q - pqR_1 + q \sum_{j=2}^N a_j R_j + qa_{N+1}cR_{N+1} + \frac{p\sigma_2}{3c^2} - p \sum_{j=1}^N d_j S_j. \quad (7.22)$$

All certificates using this formula take $N = 8$.

A first compression correction. Write $x = \cos^2 \phi$, where $\nu = (\sqrt{1-z^2} \cos \phi, \sqrt{1-z^2} \sin \phi, z)$. The determinant of the compression is

$$D = -m + d_1 x, \quad d_1 = c(b+1)(1-z^2). \quad (7.23)$$

View the compression as $\text{diag}(b, m) - (b+1)ww^T$. If its larger eigenvalue is $\mu = m - \ell$, the two-dimensional characteristic equation gives

$$\ell\{k(1-z^2) + (b+1)|w|^2 - \ell\} = (b+1)kz^2(1-z^2)x.$$

Since $|w|^2 \leq 1$, it follows that

$$\ell \geq Kx, \quad K = \frac{(b+1)kz^2(1-z^2)}{k(1-z^2) + b+1}, \quad L = \max\{0, \min(K, m-1)\}. \quad (7.24)$$

On $m > 1$, the actual norm is therefore at most $m - Lx$. Integrating (7.7) over the feasible norm interval, and using $r^{p-3} \geq c^{p-3}$ there, gives

$$\frac{2(p-1)}{c^p} \Phi_p(H) \leq U_{p,8} - \frac{p(2-p)}{c^3} \int_0^1 J(z) dz, \quad (7.25)$$

with

$$J = L \left\{ \frac{m^2}{2} + \frac{(1-3b)m}{8} + \frac{3b(1+b)z^2}{8} - \frac{3mL}{8} + \frac{5L^2}{48} \right\}. \quad (7.26)$$

Indeed $\mathbb{E}x = 1/2$, $\mathbb{E}x^2 = 3/8$, and $\mathbb{E}x^3 = 5/16$. This J is nonnegative: when $L > 0$, use $m \geq 1$, $L \leq m-1$, and $b \leq 1$ to bound the braces below by $m(m+1)/8$.

A two-sided compression correction. Keeping $|w|^2 = 1 - (1 - z^2)x$ in the same characteristic equation gives

$$\ell \geq \frac{Kx}{1 - Rx}, \quad R = \frac{(b+1)(1 - z^2)}{k(1 - z^2) + b + 1}.$$

The exceptional zero denominator is handled by continuity. The matrix inequality $-H \leq -bI + (b+1)e_1e_1^T$ also gives $-\lambda \leq 1 - Bx$, where $B = (b+1)(1 - z^2)$ and λ is the smaller compressed eigenvalue. For an integer $r \geq 1$ set

$$\begin{aligned} W &= B + M_0 - 1, \quad \Delta(x) = \sum_{j=1}^r \alpha_j x^j, \\ \alpha_j &= \min \left\{ KR^{j-1}, \max \left(0, W - \sum_{i < j} \alpha_i \right) \right\}, \quad 1 \leq j \leq r. \end{aligned} \tag{7.27}$$

Then $\Delta \leq Kx/(1 - Rx)$ and $\Delta \leq xW \leq Bx + M_0 - 1$, so both eigenvalue bounds imply $M \leq M_0 - \Delta$. Formula (7.25) now holds with

$$J = (M_0^2 + m)\mathbb{E}\Delta - d_1\mathbb{E}(x\Delta) - M_0\mathbb{E}\Delta^2 + \frac{1}{3}\mathbb{E}\Delta^3 \geq 0. \tag{7.28}$$

Nonnegativity follows directly from the feasible-interval integral of $r^2 - D$. Its angular moments are rational:

$$\mathbb{E}x^j = \mu_j := 4^{-j} \binom{2j}{j}. \tag{7.29}$$

Exact bounds on a parameter box. For clarity we give the interval rules used to turn the preceding inequalities into finite rational certificates. Write a parameter box as $[p_l, p_h] \times [b_l, b_h] \times [c_l, c_h]$. The moments $M_j, R_j, H_j, R_j - H_j$ increase in c and decrease in b ; this follows from their integral formulas with nonnegative weights. The negative ratio $(bc - b - c)/c^2$ increases in both b, c . For $p \geq 33/20$, the coefficients $p(2 - p)$, $(2 - p)a_j(p)$, and $pd_j(2 - p)$ decrease in p . For the first positive-binomial coefficient this follows from $[p(p-1)(2-p)]' = -3p^2 + 6p - 2 < 0$ for $p > 1 + 1/\sqrt{3}$; the recurrences add positive decreasing factors.

Use lower moments at (b_h, c_l) and upper moments at (b_l, c_h) . If $R_{j,l}, H_{j,l}, I_{j,l}$ are lower bounds for $R_j, H_j, R_j - H_j$, put

$$S_{j,l} = \frac{1 - b_h}{2c_h} I_{j,l} + \frac{b_l}{c_h^2} H_{j,l}.$$

The box upper bound for (7.22) is

$$\begin{aligned} \mathcal{U} &= (2 - p_l) - p_h(2 - p_h)R_{1,l} + (2 - p_l) \sum_{j=2}^8 a_j(p_l)R_{j,h} \\ &\quad + (2 - p_l)a_9(p_l)c_h R_{9,h} + \frac{p_l(b_h c_h - b_h - c_h)}{3c_h^2} - \sum_{j=1}^8 p_h d_j(2 - p_h)S_{j,l}. \end{aligned} \tag{7.30}$$

All square roots in these moments are enclosed at denominator 2^{44} by integer square roots and exact squaring. For example, if $t_l \leq t \leq t_h$, a lower bound for $R_j - H_j$ is

$$(d/c)^j \left\{ \frac{2}{3} - \frac{2j}{2j+1} t_h + \frac{2j}{3(2j+3)} t_l^3 \right\};$$

negative lower bounds are replaced by zero. The box bound for (7.19) is similarly

$$(2 - p_l) - \frac{p_h(2 - p_h)M_{1,l}}{c_h} + \frac{p_l(p_l - 1)(2 - p_l)M_{2,h}}{2P_l} + \frac{p_l(b_h c_h - b_h - c_h)}{3c_h^2}, \quad (7.31)$$

where $P_l \leq c_l^{9/5}$ is an integer fifth-root enclosure of c_l^9 .

For the correction integrals divide $[0, 1]$ into equal rational z -intervals. On a slab let $s_l = z_l^2$, $s_h = z_h^2$ and set

$$\begin{aligned} m_l &= c_l(1 - s_h) + b_l s_h, & m_h &= c_h(1 - s_l) + b_h s_l, \\ P_l &= (c_l - b_h)(1 - s_h), & P_h &= (c_h - b_l)(1 - s_l), \\ Q_l &= b_l + 1, & Q_h &= b_h + 1, \\ K_l &= s_l P_l Q_l / (P_l + Q_l), & K_h &= s_h P_h Q_h / (P_h + Q_h). \end{aligned}$$

The harmonic product is increasing in its nonnegative arguments, so these are valid bounds. For (7.26) use $L_l = \max(0, \min(K_l, m_l - 1))$ and the corresponding upper bound L_h . If $L_l = 0$ use zero; otherwise a slab lower bound is

$$\max \left(0, L_l \left\{ \frac{m_l^2}{2} + \frac{(1 - 3b_h)\hat{m}}{8} + \frac{3b_l(1 + b_l)s_l}{8} - \frac{3m_h L_h}{8} + \frac{5L_l^2}{48} \right\} \right), \quad (7.32)$$

where $\hat{m} = m_h$ if $1 - 3b_h < 0$ and $\hat{m} = m_l$ otherwise.

For (7.28) take

$$\begin{aligned} R_l &= \frac{Q_l(1 - s_h)}{(c_h - b_l)(1 - s_h) + Q_l}, & R_h &= \frac{Q_h(1 - s_l)}{(c_l - b_h)(1 - s_l) + Q_h}, \\ W_l &= Q_l(1 - s_h) + \max(0, m_l - 1), & W_h &= Q_h(1 - s_l) + \max(0, m_h - 1), \end{aligned}$$

and enclose the greedy coefficients recursively by

$$\begin{aligned} \alpha_{j,l} &= \min\{K_l R_l^{j-1}, \max(0, W_l - \sum_{i < j} \alpha_{i,h})\}, \\ \alpha_{j,h} &= \min\{K_h R_h^{j-1}, \max(0, W_h - \sum_{i < j} \alpha_{i,l})\}. \end{aligned}$$

Put $M_l = \max(1, m_l)$, $M_h = \max(1, m_h)$ and $d_{1,h} = c_h(b_h + 1)(1 - s_l)$. All coefficients are nonnegative. For degree three, evaluate the positive linear and cubic terms of (7.28) with the lower coefficients and the negative linear and quadratic terms with the upper coefficients, using the exact moments (7.29). More explicitly the lower bound is

$$(M_l^2 + m_l) \sum_j \mu_j \alpha_{j,l} - d_{1,h} \sum_j \mu_{j+1} \alpha_{j,h} - M_h \sum_{i,j} \mu_{i+j} \alpha_{i,h} \alpha_{j,h} + \frac{1}{3} \sum_{i,j,k} \mu_{i+j+k} \alpha_{i,l} \alpha_{j,l} \alpha_{k,l}.$$

For degree six use the sharper lower bound

$$\sum_j \max\{0, (M_l^2 + m_l) \mu_j - d_{1,h} \mu_{j+1}\} \alpha_{j,l} - M_h \sum_{i,j} \mu_{i+j} \alpha_{i,h} \alpha_{j,h} + \frac{1}{3} \left(\sum_j \mu_j \alpha_{j,l} \right)^3. \quad (7.33)$$

Clipping the linear coefficients is valid because $d_1 \leq (b + 1)m \leq 2m$, $M_0^2 \geq m$, and $\mu_j \geq \mu_{j+1}$. The last term uses Jensen's inequality. Replace either resulting lower bound by its maximum with zero,

since $J \geq 0$, and round it down to the grid 2^{-80} . Summing the slab bounds times their lengths gives $G_l \leq \int J$. The final box bound is

$$\mathcal{U} - \frac{p_h(2 - p_h)}{c_h^3} G_l. \quad (7.34)$$

The degree-three and degree-six implementations may accept a box immediately if $\mathcal{U} < 0$, without computing a correction.

Parameter interval	Upper bound	z slabs	Accepted boxes
$[9/5, 19/10]$	(7.31)	—	373
$[7/4, 9/5]$	(7.30)	—	377
$[17/10, 7/4]$	(7.32), (7.34)	32	2202
$[5/3, 17/10]$	degree-three correction	128	9853
$[33/20, 5/3]$	degree-six correction	128	19354

Table 1: Exact certificates for $0 \leq b \leq 1 \leq c$, $b + c \leq 2p/(3 - p)$. Every accepted upper bound is strictly negative. All five certificates have complete coverage and zero unresolved boxes.

7.5 The negative-middle sector and verification of coverage

If $-1 \leq b \leq 0$ and $1 \leq b + c$, then $c \geq 1$. The two terms in $\overline{D}$ have opposite signs. Using $c^{p-2} \leq M_0^{p-2} \leq 1$ separately on them gives

$$\int_0^1 \overline{D} M_0^{p-2} dz \leq -\frac{(1-b)c^{p-1}}{3} - \frac{b}{3}.$$

Together with (7.18), this proves

$$\frac{2(p-1)}{c^p} \Phi_p(H) \leq (2-p) - \frac{p(2-p)M_1}{c} + \frac{p(p-1)(2-p)M_2}{2c^p} - \frac{p(1-b)}{3c} - \frac{pb}{3c^p}. \quad (7.35)$$

For a band $[a, h] \subset [33/20, 9/5]$, use an exact lower bound $v_l \leq c_l^a$. On a parameter box the upper bound is

$$(2 - p_l) - \frac{p_h(2 - p_h)M_{1,l}}{c_h} + \frac{p_l(p_l - 1)(2 - p_l)M_{2,h}}{2v_l} - \frac{p_l(1 - b_h)}{3c_h} - \frac{p_h b_l}{3v_l}. \quad (7.36)$$

Only positive terms used the weaker denominator c^a . The coefficient monotonicities and moment endpoint choices are the ones proved above. If $a = u/v$ is rational, c^a is enclosed by the integer v -th root of c^u , with denominator 2^{40} and exact power checks. For the first row below the whole parameter interval is used in each box, so its leaves have only b, c coordinates.

Parameter interval $[a, h]$	Range of $b + c$	Root degree	Accepted boxes
$[7/4, 9/5]$	$[1, 3]$	4	42
$[17/10, 7/4]$	$[1, 3]$	10	437
$[5/3, 17/10]$	$[1, 34/13]$	3	360
$[33/20, 5/3]$	$[1, 5/2]$	20	568

Table 2: Exact certificates for $-1 \leq b \leq 0$, using (7.36). The stated sum caps contain $b + c \leq 2p/(3 - p)$ throughout each band. All bounds are strictly negative.

Finite verification protocol. The supplement contains the complete rational leaf lists, the producers, and independent verifiers. A rational number is stored as its integer numerator and positive denominator. Acceptance uses exact rational arithmetic; no floating-point comparison accepts a box. Root enclosures are verified by integer powers. For example, at denominator Q a lower enclosure of $x^{1/r}$ is k/Q , where

$$k^r \leq xQ^r < (k+1)^r.$$

This is computed by integer root search after clearing denominators.

For Table 1, the initial rectangle is the parameter band times $[0, 1] \times [1, C]$, with $C = 38/11$ in the first row and $C = 3$ in the others. The only trimming rules are

$$b_h \leftarrow \min(b_h, S(p_h) - c_l), \quad c_h \leftarrow \min(c_h, S(p_h) - b_l), \quad p_l \leftarrow \max\left(p_l, \frac{3(b_l + c_l)}{b_l + c_l + 2}\right). \quad (7.37)$$

A box with $b_l + c_l > S(p_h)$ is empty of feasible points. Each rule preserves every feasible point because $S(p)$ is increasing. Otherwise bisect a rational coordinate interval until a strictly negative box bound is obtained. The negative-middle certificates start from their parameter band times $[-1, 0] \times [1, C+1]$, where C is the sum cap in Table 2, and trim only by $1 \leq b + c \leq C$. These are necessary linear inequalities.

Besides recomputing every leaf bound, the independent verifiers check coverage without trusting the producer's traversal. Collect all rational leaf endpoints in each coordinate and form their elementary open cells. On each such cell, inclusion in any particular leaf is constant. For the curved domain a cell meets its interior precisely when its lower $b + c$ corner is smaller than S at its upper p endpoint; empty cells are omitted. For a fixed sum cap, use the corresponding two linear tests. Exact rational comparisons verify that every remaining cell is contained in at least one leaf. Finite unions of closed leaves are closed, and the target domain is the closure of its interior, so this also covers all faces and endpoints. The verifiers implement this arrangement with sweeps and integer masks rather than materializing every cell. In particular the last positive-middle certificate covers 14 555 358 398 represented cells, checks 25 501 root enclosures, and exactly reproduces all 19 354 negative leaf bounds.

Here are the supplement filenames defining the nine certificate packets. For each stem in the following display, the files ending in `_certificate.py`, `_leaves.jsonl.gz`, and `_result.json` are included:

```
session2_parametric_cone, session2_binomial_cone, session2_gap_cone, session2_ser
ies_cone, session2_series_33_cone;
session2_negative_middle, session2_negative_middle_17, session2_negative_middle
_53, session2_negative_middle_33.
```

The five positive packets also include their `_verify.py` and `_verify_result.json` files. The negative packets have independent checkers `session2_independent_negative_check.py`, `session2_independent_negative_17_check.py`, `session2_independent_negative_53_check.py`, and `session2_independent_negative_33_check.py`, with matching JSON results. The data are explicit companion inputs to this computer-assisted proof; the thousands of leaves are not asserted to be printed here. The algorithms and inequalities above specify the mathematical obligations checked by these files.

Completion of Proposition 7.3. For $p_0 \in [33/20, 19/10]$, the five rows of Table 1 cover $0 \leq b \leq 1 \leq c$. The four rows of Table 2 cover $b \leq 0$ below $9/5$, and the earlier elementary bound covers $b \leq 0$ above it. The case $c \leq 1$ was already disposed of. Finally (7.14) covers $p_0 \in [\rho, 2)$, overlapping the certified interval. These sectors exhaust (7.6) with $b \leq 1$. $\square$

7.6 A direct certificate close to isotropy

The preceding parameter bands do not cover the whole interval down to $p_0 = 3/2$. A different finite certificate proves a complementary result.

Proposition 7.4 (Computer-assisted near-isotropic sign). *For every $-1 \leq b \leq c$ with $1 \leq b + c \leq 243/128$,*

$$\Phi_{3/2}(\text{diag}(-1, b, c)) < 0.$$

Proof. We give the complete interval-integration rule, whose accepted rectangles are supplied in the supplement. Put

$$w_3 = z^2, \quad w_1 = \frac{1 - z^2}{1 + t^2}, \quad w_2 = \frac{(1 - z^2)t^2}{1 + t^2}, \quad 0 \leq z, t \leq 1.$$

Here $t = \tan \phi$ describes half of a quadrant; exchanging w_1, w_2 describes the other half. Sign changes of the normal do not affect the compression invariants. Consequently

$$\Phi_{3/2}(H) = \frac{2}{\pi} \int_0^1 \int_0^1 \frac{\mathcal{B}(w_1, w_2, w_3) + \mathcal{B}(w_2, w_1, w_3)}{1 + t^2} dt dz. \quad (7.38)$$

The trace, determinant, norm, and integrand in this formula are

$$\begin{aligned} \tau &= -(1 - w_1) + b(1 - w_2) + c(1 - w_3), \\ D &= bcw_1 - cw_2 - bw_3, \quad M = \frac{1}{2}\{|\tau| + \sqrt{\tau^2 - 4D}\}, \\ \mathcal{B} &= \frac{M^2 + 3D}{2\sqrt{M}}, \quad \mathcal{B}(0) = 0. \end{aligned} \quad (7.39)$$

The positive prefactor $2/\pi$ is irrelevant to the sign, so no irrational integration weight needs an enclosure.

All endpoints and arithmetic bounds use integer multiples of Q^{-1} , $Q = 2^{40}$, with outward rounding. Addition and subtraction are exact; multiplication and division use the integer floor or ceiling after clearing the positive denominator. Square roots use integers r satisfying $r^2 \leq aQ < (r + 1)^2$ for an input a/Q . The independent verifier uses an integer-only root search and verifies both square inequalities. The independent C++ replay also checks that the absolute value of every stored scalar is at most 2^{55} . Products are then bounded by 2^{110} and the angular sums by 2^{73} , within signed 128-bit arithmetic. Thus machine overflow is excluded by explicit checked bounds, rather than assumed from the use of integer arithmetic.

On each parameter rectangle and each angular rectangle, interval arithmetic gives $\tau_l \leq \tau \leq \tau_h$ and $D_l \leq D \leq D_h$. Set $A = \max(|\tau_l|, |\tau_h|)$ and round upwards to obtain

$$M_h \geq \frac{1}{2}\{A + \sqrt{\max(0, A^2 - 4D_l)}\} \geq M.$$

For the actual determinant first use (7.7), then the linear monotonicity in the determinant:

$$\mathcal{B}(M, D) \leq \mathcal{B}(M_h, D) \leq \frac{M_h^2 + 3D_h}{2\sqrt{M_h}}. \quad (7.40)$$

There is no assumption that D_h is itself feasible at M_h . Round the numerator upwards. If it is nonnegative use a lower square-root denominator; if it is negative use an upper denominator. Round the quotient upwards. The zero-norm case is evaluated directly. Add the two upper bounds for the

exchanged weights in (7.38). Multiply a nonnegative sum by the upper bound on $(1 + t^2)^{-1}$, and a negative sum by its lower bound. Summation over the N^2 equal-area angular rectangles, divided by N^2 with upward rounding, is a rigorous upper bound on the double integral. The certificate uses $N \in \{32, 64, 128, 256, 512\}$.

The initial coefficient rectangle is

$$-1 \leq b \leq \frac{243}{256}, \quad \frac{1}{2} \leq c \leq \frac{371}{128}.$$

It contains the required domain. Trimming uses only $b \leq c$ and $1 \leq b + c \leq 243/128$, followed by binary subdivision of a coefficient interval. An interval that cannot be subdivided and has not obtained a negative bound is a failure, never an accepted leaf. The complete packet has 3748 accepted rectangles, 3747 binary splits, and no discarded or unresolved rectangles. Its largest accepted upper bound for the double integral is the negative rational number

$$-\frac{7472810}{1099511627776}. \quad (7.41)$$

Every leaf bound was independently reproduced with integer arithmetic; 1 450 614 784 root-enclosure checks passed. An independent arrangement of the leaf edges covered all 130 374 cells meeting the coefficient polygon, with every leaf used. More explicitly, an open rectangle $(b_l, b_h) \times (c_l, c_h)$ meets the polygon interior exactly when

$$\max\{b_l, 1 - c_h\} < \min\{b_h, c_h, S/2, S - c_l\}, \quad S = 243/128.$$

This is the exact intersection test used by the coverage checker. Closed coverage includes its edges, as above.

The source is `session2_near_iso_angular.cpp`. The explicit input, leaf, and traversal files are in `near_iso_243/`: `session2_near_iso_angular_result.txt`, `session2_near_iso_angular_leaves.tsv`, and `session2_near_iso_angular_tree.tsv`. The independent verifier and coverage algorithm are `session2_near_iso_243_verify.cpp` and `session2_near_iso_243_coverage.py`; their results are the `_replay.json` and `_coverage.json` files in that directory. This finite computation proves the claimed strict sign by (7.38)–(7.40). $\square$

7.7 Consequences for the original maximum estimate

Proof of Theorem 7.1. For $1/6 < \delta \leq 9/40$ we have $33/20 \leq p_0 < 2$. Proposition 7.3 gives strict negativity of Φ_{p_0} on the compact part of the normalized null cone with $b \leq 1$. It persists for exponents $q > p_0$ sufficiently close to p_0 . Lemma 7.2 supplies strict negativity on the rest of that cone for every $p_0 < q < 2$. Given any requested $p > p_0$, choose such a $q < \min(p, 2)$. Compactness and homogeneity give uniform strict null-cone negativity at exponent q . Equations (7.2)–(7.3) and Theorem 5.1 prove the original estimate at q , with exactly the original measurable coefficients and interior C^2 regularity. Hölder’s inequality on B_1 gives the estimate at p . The range $\delta \leq 1/6$ follows from Theorem 4.1. Theorem 2.1 proves failure at and below the claimed threshold throughout these ranges.

For $\delta_* = 128/499$, (7.4) gives $S = 243/128$. Proposition 7.4 and compactness imply strict negativity of Φ_q on this same cone for every $q > 3/2$ sufficiently close to $3/2$. For any given $p > 3/2$, choose such a $q < p$. The transfer theorem and Hölder give the estimate. Increasing δ only shrinks the coefficient class, so this proves the whole interval $\delta_* \leq \delta < 1/3$. Here $\delta_* > 1/4$, and the target threshold is $3/2$. The necessity theorem excludes $p = 3/2$ and all smaller exponents. Strict matrix

negativity at $3/2$ has not been used to infer a maximum estimate at the excluded Newton-potential endpoint.

Finally fix $\delta > 9/40$ and set $p_* = 33/20$. The normalized null cone now has the strict sum cap $b + c \leq 1/\delta - 2 < 22/9 = 2p_*/(3 - p_*)$. On $b \leq 1$, Proposition 7.3 at p_* already gives strict negativity on the larger cone. On $b \geq 1$, the concavity part of Lemma 7.2, used at $p = p_0 = p_*$, and the strict sum cap give

$$\Phi_{p_*}(H) \leq \frac{[-p_* + (3 - p_*)m]m^{p_*-1}}{3(p_* - 1)} < 0.$$

Since $p_* > 3/2$, Theorem 5.1 applies directly at p_* , and Hölder gives all $p \geq p_*$. This does not include the failing pair $\delta = 9/40, p = 33/20$. $\square$

7.8 An additional explicit sufficient condition

The following independent estimate covers additional points and also settles $p = 2$ whenever $\delta > 1/6$. It is a sufficient criterion, not an identification of the sharp operator norm.

Proposition 7.5 (Quantitative perturbation). *Let $n = 3$, $0 < \delta < 1/3$ and $3/2 < p < 2$. If there is $1 < r < p$ such that*

$$\kappa := 2(1 - 3\delta) \left[\frac{\sqrt{5/2}}{r - 1} \right]^\theta < 1, \quad \theta = \frac{1/p - 1/2}{1/r - 1/2}, \quad (7.42)$$

then $\text{MP}(3, \delta, p)$ holds. It also holds at $p = 2$ whenever $\delta > 1/6$.

Proof. Write $T = D^2\Delta^{-1}$ on $\mathbb{R}^3$ and $S = T - I/3$, where $(I/3)f = fI/3$. We first prove the complex-valued operator estimate

$$\|S\|_{L^p \rightarrow L^p(\text{Sym}_3)} \leq \sqrt{\frac{2}{3}} \left[\frac{\sqrt{5/2}}{r - 1} \right]^\theta. \quad (7.43)$$

The matrix norm is Frobenius. Plancherel gives $\|S\|_{2 \rightarrow 2} = \sqrt{2/3}$, because its symbol is $\xi \otimes \xi / |\xi|^2 - I/3$.

For completeness, the martingale input used to bound S on L^r is the following Hilbert-valued differential-subordination inequality: if a Hilbert-valued martingale N starts at zero and is differentially subordinate to a Hilbert-valued martingale M , then

$$\|N\|_{L^r} \leq (r^* - 1)\|M\|_{L^r}, \quad r^* = \max\{r, r/(r - 1)\}.$$

Here differential subordination means that the difference of scalar quadratic variations $[M] - [N]$ is nonnegative and nondecreasing, with the initial domination included; the norms may be taken over bounded stopping times. This theorem, in the precise Hilbert-valued form just used, is recalled in [9, Theorem 1.1]. For a martingale closed by a fixed terminal variable, conditional Jensen identifies its stopping-time norm with the terminal L^r norm.

Let B_t be Brownian motion with generator $\Delta/2$ and $P_t = e^{t\Delta/2}$. Set

$$D^{ij} = \frac{1}{2}(e_i e_j^t + e_j e_i^t) - \frac{\delta_{ij}}{3} I.$$

For every real or complex vector z , direct summation yields

$$\sum_{i,j=1}^3 |D^{ij}z|^2 = \frac{5}{3}|z|^2. \quad (7.44)$$

Fix $t_0 > 0$, $x \in \mathbb{R}^3$ and smooth compactly supported complex f . Consider

$$M_s = P_{t_0-s} f(x + B_s), \quad N_s^{ij} = \int_0^s D^{ij} \nabla P_{t_0-t} f(x + B_t) \cdot dB_t.$$

By (7.44), N is differentially subordinate to $\sqrt{5/3}M$. Realifying complex vectors preserves all these Hilbert norms. The martingale theorem and integration over x therefore bound the terminal tensor norm by $\sqrt{5/3}(r^* - 1)\|f\|_r$.

To identify its projected multiplier, pair each terminal component with $\overline{g_{ij}(x + B_{t_0})}$, sum over i, j , and integrate over x . Itô isometry and invariance of Lebesgue measure give

$$\sum_{i,j} \int_0^{t_0} \int_{\mathbb{R}^3} D^{ij} \nabla P_t f \cdot \overline{\nabla P_t g_{ij}} \, dx \, dt.$$

Hölder bounds this form by the preceding constant times $\|f\|_r \|(g_{ij})\|_{r'}$. Its multiplier is

$$\left(\frac{\xi_i \xi_j}{|\xi|^2} - \frac{\delta_{ij}}{3} \right) (1 - e^{-t_0 |\xi|^2}).$$

Letting $t_0 \rightarrow \infty$ in smooth Fourier pairings, then using duality and density, proves $\|S\|_{r \rightarrow r} \leq \sqrt{5/3}(r^* - 1)$. Complex interpolation with the exact L^2 norm proves (7.43) for $1 < r < p < 2$.

Put $\beta = 1 - 3\delta$ and $E = A - I/3$. The extreme eigenvalue vectors of the trace-one coefficient simplex give $|E| \leq \beta\sqrt{2/3}$. Since Sh is trace free,

$$A : Th = h/3 + E : Sh.$$

Thus the equation $A : Th = f$ is equivalent to $h = 3f - 3E : Sh$. By (7.43), its second term has operator norm at most κ . A Neumann series gives

$$\|h\|_p \leq \frac{3}{1 - \kappa} \|f\|_p. \quad (7.45)$$

For smooth A equal to $I/3$ outside B_2 and $0 \leq g \in C_c^\infty(B_2)$, take $f = -g$. Then the solution h is supported in $\overline{B_2}$. The positive Newton barrier and the inner-ball approximation in the proof of Theorem 5.1 use only such an inverse estimate; applying that argument with (7.45) proves the claim in exactly the original coefficient and function classes. At $p = 2$, the contraction factor is 2β , which is less than one exactly when $\delta > 1/6$; the required Newton condition $p > 3/2$ still holds. $\square$

7.9 A precise conditional route through positive sources

The remaining question can be approached through a more specific inequality than an unrestricted sharp Hessian-transform norm estimate. The next conjecture is recorded as an unproved sufficient mechanism.

Conjecture 7.6 (Positive-source spectral pairing). *For $3/2 < p < 2$, put $C_p = (3 - p)/[3(p - 1)]$. For every nonnegative $g \in C_c^\infty(\mathbb{R}^3)$,*

$$\int_{\mathbb{R}^3} g^{p-1} \lambda_{\min}(Tg) \, dx \geq -C_p \int_{\mathbb{R}^3} g^p \, dx, \quad T = D^2 \Delta^{-1}. \quad (7.46)$$

The Hessian of the Newton potential uses the kernel $-1/(4\pi|x|)$, so $\text{tr}(Tg) = g$. Boundedness of T on L^p , the Lipschitz continuity of the smallest eigenvalue, and continuity of $g \mapsto g^{p-1}$ from nonnegative L^p to $L^{p'}$ show that (7.46) is equivalent to its formulation for all nonnegative $g \in L^p(\mathbb{R}^3)$.

Proposition 7.7 (Conditional implication). *If Conjecture 7.6 holds at a fixed p , then $\text{MP}(3, \delta, p)$ holds for every $0 < \delta < 1/3$ satisfying $p > 3(1 - 2\delta)$.*

Proof. Let $q = p/(p-1) \in (2, 3)$, and take a smooth compactly supported matrix field $M \geq \delta(\text{tr } M)I$. Write

$$w = \text{tr } M, \quad M = \delta w I + \beta W, \quad \beta = 1 - 3\delta, \quad W \geq 0, \quad \text{tr } W = w, \quad \nu = \text{Div}^2 M.$$

Set $h = \Delta^{-1}\nu$. Then $h = \delta w + \beta \sum_{i,j} T_{ij} W_{ij}$. Test this identity against $g = w^{q-1}$ and use self-adjointness of the scalar Riesz multipliers. Because $g^{p-1} = w$ and $g^p = w^q$, the conjectured inequality gives

$$\begin{aligned} \int hg &= \delta \int w^q + \beta \int W : Tg \\ &\geq \delta \int w^q + \beta \int w \lambda_{\min}(Tg) \geq (\delta - \beta C_p) \int w^q. \end{aligned}$$

The coefficient is exactly

$$\delta - \beta C_p = \frac{p-3+6\delta}{3(p-1)} > 0.$$

If the support of M lies in a fixed ball, then

$$\|\Delta^{-1}\nu\|_q \leq C_q \|\nu\|_{\text{TV}}. \quad (7.47)$$

Near that ball, this follows from Minkowski and local L^q integrability of $|x|^{-1}$ for $q < 3$. At infinity, ν has zero mass and zero first moments, since it is a double divergence of a compactly supported field. Subtract the constant and linear Taylor terms of the Newton kernel; its remaining convolution is bounded by $C|x|^{-3}\|\nu\|_{\text{TV}}$, which is in L^q at infinity. Hölder and (7.47) now give

$$\|\text{tr } M\|_q \leq C_{\delta,p} \|\text{Div}^2 M\|_{\text{TV}}.$$

Lemma 4.6 extends this estimate to cone-valued matrix measures, and Proposition 4.8 gives the original Dirichlet estimate. These two statements require only $q > 1$; the restriction $q < 2$ in the earlier Mellin construction is not part of either transfer. This proves the assertion. $\square$

Proposition 7.8 (Proved cases and sharp constant). *The inequality (7.46) holds at $p = 2$ for every nonnegative L^2 source, with $C_2 = 1/3$. For $3/2 < p \leq 2$, it holds for every nonnegative radial $g \in L^p(\mathbb{R}^3)$, with no monotonicity assumption. The constant C_p is optimal within this radial class.*

Proof. At $p = 2$, write $Sg = Tg - (g/3)I$. For trace-free $H \in \text{Sym}_3$, $\lambda_{\min}(H) \geq -\sqrt{2/3}|H|$. Thus Plancherel and Cauchy–Schwarz give

$$\int g \lambda_{\min}(Tg) \geq \frac{1}{3} \|g\|_2^2 - \sqrt{2/3} \|g\|_2 \|Sg\|_2 = -\frac{1}{3} \|g\|_2^2.$$

For the radial assertion, initially suppose g is bounded and compactly supported, put $t = r^3$, $f(t) = g(t^{1/3})$, and define $H(t) = t^{-1} \int_0^t f(s) \, ds$. The radial and tangential eigenvalues of Tg are $f - 2H/3$ and $H/3$, respectively. Where $H > 0$, write $z = f/H$. Apart from the positive polar factor $4\pi/3$, the deficit in (7.46) equals

$$\int_0^\infty H^p V(z) \, dt, \quad V(z) = z^{p-1} \min\{z - 2/3, 1/3\} + C_p z^p.$$

Let $z_0 = (p-1)/p$ and $D = 2z_0^{p-2}/(3p)$. We claim

$$V(z) \geq D[pz - (p-1)] \quad (z \geq 0). \quad (7.48)$$

On $0 \leq z \leq 1$, one has $V(z) = (1 + C_p)z^p - (2/3)z^{p-1}$, a convex function since

$$V''(z) = (p-1)z^{p-3}[p(1 + C_p)z - \frac{2}{3}(p-2)] \geq 0.$$

Its value and derivative at z_0 are zero and Dp , so the right side of (7.48) is its supporting tangent. On $z \geq 1$ the other branch is $V(z) = C_p z^p + z^{p-1}/3$, also convex because $pC_p z + (p-2)/3 > 0$. Its right derivative at one is $(p+1)/[3(p-1)] \geq Dp$. Indeed, for $x = p/(p-1) \in [2, 3]$,

$$2x - 1 > 2\sqrt{x} \geq 2x^{(x-2)/(x-1)}.$$

The two branches have the same value at one, so the supporting-line inequality holds also on $z \geq 1$.

Since $(tH^p)' = H^p[pz - (p-1)]$ and tH^p tends to zero at both endpoints, integrating (7.48) proves the result. An initial interval on which $H = 0$ contributes nothing; boundedness near zero and compact support of f justify both endpoint limits. Nonnegative radial L^p approximation proves the general assertion.

Finally set $\alpha = 3/p$ and, for $0 < \varepsilon < 1$, let

$$g_\varepsilon(r) = \begin{cases} \varepsilon^{-\alpha}, & 0 \leq r \leq \varepsilon, \\ r^{-\alpha}, & \varepsilon < r \leq 1, \\ 0, & r > 1. \end{cases}$$

For $\varphi_\varepsilon = \Delta^{-1}g_\varepsilon$ and $\varepsilon < r < 1$, direct radial integration gives

$$\begin{aligned} \varphi'_\varepsilon(r)/r &= \frac{r^{-\alpha}}{3-\alpha} - \frac{\alpha\varepsilon^{3-\alpha}}{3(3-\alpha)r^3}, \\ \varphi''_\varepsilon(r) &= \frac{1-\alpha}{3-\alpha}r^{-\alpha} + \frac{2\alpha\varepsilon^{3-\alpha}}{3(3-\alpha)r^3}. \end{aligned}$$

The second is the smaller eigenvalue on this annulus. Since $\alpha p = 3$ and $\alpha(p-1) = 3-\alpha$,

$$\int g_\varepsilon^p = 4\pi[\log(1/\varepsilon) + 1/3], \quad \int g_\varepsilon^{p-1} \lambda_{\min}(Tg_\varepsilon) = 4\pi \frac{1-\alpha}{3-\alpha} \log(1/\varepsilon) + O(1).$$

The core and the ε -dependent annular correction contribute only $O(1)$. Their quotient tends to $-(\alpha-1)/(3-\alpha) = -C_p$. Smooth nonnegative radial approximation preserves this limiting conclusion, proving optimality. $\square$

7.10 The remaining obligation

Combining the preceding results with Theorem 2.1, the unresolved part of (1.2) is contained in

$$\boxed{n = 3, \quad \frac{9}{40} < \delta < \frac{128}{499}, \quad \max\{3/2, 3(1-2\delta)\} < p < \frac{33}{20}.} \quad (7.49)$$

Points satisfying (7.42) are already covered and are removed from this remaining obligation. In particular, (7.49) is an outer description of the outstanding parameter set, not a claim that every point in the displayed strip lacks a proof.

The positive-source inequality in Conjecture 7.6, if established throughout $3/2 < p < 2$, would close the gap by Proposition 7.7. Its proved radial case and its proved $p = 2$ case do not establish the general inequality. Thus the results above resolve the exact question in the stated dimensions and parameter ranges, while leaving the unrestricted three-dimensional assertion open within this manuscript.

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